

14. A bag contains 5 white, 7 red and 8 black balls. If four balls are drawn one by one without replacement find the probability of getting all white balls.

15. An urn contains 5 white and 8 black balls two successive drawing of three balls at a time without replacement such that the balls are not replaced before the second draw. Find the probability that the first draw gives 3 white balls and second draw gives 3 black balls.

16. A coin is tossed twice and the four possible outcome are assumed to be equally likely. If E is the event "both head and tail have occurred" and F the event "at most one tail has occurred". Find $P(E|F)$ and $P(F|E)$.

17. An urn contains 10 white and 3 black balls while another urn contains 3 white and 5 black balls. Two are drawn from first urn and put into the second urn and then a ball is drawn from the second urn. Find the probability that it is white ball.

18. There are two bags containing 5 red, 7 white and 3 red, 12 white balls respectively. A ball is drawn from one of the two bags, find the probability of drawing a red ball.

19. A purse contains 2 silver and 4 copper coins. A second purse contains 4 silver and 3 copper coins. A coin is pulled out at random from one of the two purses. What is the probability that a silver coin is drawn?

20. A bag contains 6 red and 8 black balls and another bag contains 8 red and 6 black balls. A ball is drawn from first bag and without noticing its colour is put in the second bag. A ball is then drawn from the second bag. Find the probability that the ball drawn is red in colour.

1.5. RANDOM VARIABLE AND PROBABILITY DISTRIBUTION

If an experiment is conducted under identical conditions, values so obtained may not be same. Observations are always taken about a **factor** or **character** under study, which can take different values. This **factor** or **character** is termed as **variable**. The observations may be the number of certain objects or items or their measurements. These observations vary even though the experiment is conducted under identical conditions. Hence, we have a set of outcomes of a random experiment. A rule that assigns a real number to each outcome is called **random variable**. The rule is nothing but a function of the variable, say, X that assigns a unique value to each outcome of the random experiment. It is the value that there is a value for each outcome, which it takes with certain probability. Thus when a variable takes the value x_i with probability p_i , ($i = 1, 2, 3, \dots, n$), then X is called **random variable** or **stochastic variable** or a **variante**.

1.6. DISCRETE RANDOM VARIABLE

A random variable X , which can take only a finite number of values in an interval of the domain, is called discrete random variable.

A random variable X is said to have an exponential distribution if its probability density function is given by $f(x) = \lambda e^{-\lambda x}$, $\lambda > 0$.

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Example:

1. Number appearing on top of a die when it is thrown.
2. The number of telephone calls received per day.
3. Number of mistakes in a page.
4. Number of defective items in a lot.

1.7. DISCRETE PROBABILITY DISTRIBUTION

If a random variable x can assume a discrete set of values say $x_1, x_2, \dots, x_n$ with respect to probabilities $p_1, p_2, \dots, p_n$ such that $p_1 + p_2 + \dots + p_n = 1$ i.e., $\sum_{i=1}^n p_i = 1$ then occurrences of values x_i with respective probabilities p_i is called the discrete probability distribution of X .

For example, In a throw of a pair of dice the sum (X) is discrete random variable which is an integer between 2 and 12 with probabilities $P(X)$ given as

X	2	3	4	5	6	7	8	9	10	11	12
$P(X)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

This constitute a discrete probability distribution.

1.8. PROBABILITY FUNCTION OR PROBABILITY MASS FUNCTION (pmf)

Probability function of probability mass function (p.m.f.) of a random variable X is mathematical function $p(x)$ which gives the probabilities corresponding to different possible discrete set of values say $x_1, x_2, x_3, \dots, x_n$ of variable x .

$$i.e., \quad p(x_i) = p(x = x_i)$$

The function $p(x)$ satisfies the condition.

- (i) $p(x_i) \geq 0$
- (ii) $\sum p(x_i) = 1$

1.9. CUMULATIVE DISTRIBUTION FUNCTION (DISTRIBUTION FUNCTION)

If X is a random variable then $P(X \leq x)$ is called the cumulative distribution function (cdf) or distribution function and is denoted by $F(x)$

$$F(x) = P(X \leq x)$$

1.10. EXPECTATION OF A DISCRETE RANDOM VARIABLE

If x is discrete random variable which assumes the discrete set of values $x_1, x_2, \dots, x_n$ with respective probabilities $p_1, p_2, \dots, p_n$ then the expectation or expected value of x is denoted by $E(X)$ and defined as

$$E(X) = x_1 p_1 + x_2 p_2 + x_3 p_3 + \dots + x_n p_n$$

$$= \sum_{i=1}^n x_i p_i$$

Similarly the expected value of X^2 is defined as $E(X^2) = \sum_{i=1}^n x_i^2 p_i$

Properties:

1. If X is a random variable and ' a ' is constant then
 - (i) $E(a) = a$
 - (ii) $E(aX) = aE(X)$
 - (iii) $E(X - \mu) = 0$
2. If x and y are two random variables then $E(X \pm Y) = E(X) \pm E(Y)$.
3. $E(XY) = E(X)E(Y)$ if X and Y are two independent random variable.
4. if $y = ax + b$ where a and b are constant then $E(Y) = aE(X) + b$.

1.11. VARIANCE AND STANDARD DEVIATION OF DISCRETE RANDOM VARIABLE

The variable of discrete random variable X is expected value of $(X - \mu)^2$ where μ is mean of variable.

$$\begin{aligned} \text{Var } X &= V(X) = E(X - \mu)^2 \\ &= \sum_{i=1}^n (x_i - \mu)^2 \\ &= \sum p_i (x_i - \mu)^2 \\ &= \sum p_i x_i^2 + \sum p_i \mu^2 - 2 \sum p_i \mu x_i \\ &= \sum p_i x_i^2 + \mu^2 \sum p_i - 2\mu \sum p_i x_i \\ &= \sum p_i x_i^2 + \mu^2 - 2\mu^2 \\ &= \sum p_i x_i^2 - \mu^2 \\ &= E(X^2) - [E(X)]^2 \end{aligned}$$

The standard deviation (SD) of a random variable x is denoted by s , then

$$SD(X) = \sqrt{V(X)} = \sqrt{E(X^2) - (E(X))^2}$$

SOLVED EXAMPLES

Example 1.41. (i) A pair of two coins is tossed, what is the expected value?
(ii) A pair of dice is thrown together, find the expected value.

Solution. (i) Expected value or mean value $= E(X) = \mu$

$$= \sum_{i=1}^n p_i \cdot x_i$$

(Here X is a discrete random variable)

In tossing of two coins, probability distribution is represented in tabular form as follows:

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X	0	1	2
$P(x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

$$E(X) = \frac{1}{4} \times 0 + \frac{1}{2} \times 1 + \frac{1}{4} \times 2 = 1$$

As the probability of getting no head, one head and two heads is respectively $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$.

(i) In a throw of pair of dice the sum (X) is a discrete random variable which is an integer between 2 and 12 with the probabilities as given below:

X	2	3	4	5	6	7	8	9	10	11	12
$P(X)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

$$\begin{aligned} \therefore \text{Expected value} &= E(X) = \mu = \frac{1}{36} \cdot 2 + \frac{2}{36} \cdot 3 + \frac{3}{36} \cdot 4 + \dots + \frac{3}{36} \cdot 10 + \frac{2}{36} \cdot 11 + \frac{1}{36} \cdot 12 \\ &= \frac{252}{36} = 7 \end{aligned}$$

Note. The variance in each of the above cases is given by

$$\sigma^2 = \sum_{i=1}^n p_i x_i^2 - \mu^2 = \sum p_i x_i^2 - \mu^2$$

In the tossing of two coins, we have

$$\sum p_i x_i^2 = \frac{1}{4} \cdot (0)^2 + \frac{1}{2} \cdot (1)^2 + \frac{1}{4} \cdot (2)^2 = \frac{3}{2}$$

$$\therefore \text{Variance} = \sigma^2 = \sum p_i x_i^2 - \mu^2 = \frac{3}{2} - 1 = \frac{1}{2}$$

In case of a pair of dice, we have

$$\sum p_i x_i^2 = \frac{1}{36} \cdot 4 + \frac{2}{36} \cdot 9 + \frac{3}{36} \cdot 16 + \frac{4}{36} \cdot 25 + \frac{5}{36} \cdot 36 + \frac{6}{36} \cdot 49$$

$$= \frac{1}{36} (4 + 18 + 48 + 100 + 180 + 294 + 320 + 324 + 300 + 242 + 144)$$

$$= \frac{1}{36} (1974) = \frac{329}{6}$$

$$\therefore \text{Variance} = \sigma^2 = \frac{329}{6} - (7)^2 = \frac{35}{6}$$

$$\therefore \text{Standard deviation} = \sigma = \sqrt{\frac{35}{6}}$$

Example 1.42. A random variable X has the following distribution

X	-2	-1	0	1	2	3
$P(X)$	0.1	k	0.2	$2k$	0.3	k

Determine: (i) k , (ii) Mean, (iii) Variance.

Solution. (i) Since $\sum P(x) = 1$

$$\Rightarrow 0.1 + k + 0.2 + 2k + 0.3 + k = 1$$

$$0.6 + 4k = 1$$

$$4k = 0.4$$

$$k = 0.1$$

$\therefore$ Now the probability distribution of the random variable X is

X	-2	-1	0	1	2	3
$P(X)$	0.1	0.1	0.2	0.2	0.3	0.1

(ii) Mean = $\sum xp(x)$

$$= (-2) \times (0.1) + (-1) \times (0.1) + (0) \times (0.2) + (1) \times (0.2) + (2) \times (0.3) + (3) \times (0.1)$$

$$= 0.8$$

(iii)

$$\text{Variance} = \sum x^2 p(x) - [\sum xp(x)]^2$$

$$= (-2)^2(0.1) + (-1)^2(0.1) + (0)^2(0.2) + (1)^2(0.2) + (2)^2(0.3) + (3)^2(0.1) - (0.8)^2$$

$$= 2.16$$

Example 1.43. A bag contains 8 items of which 2 are defective. A man selects 3 items at random. Find the expected number of defective items he has drawn.

Solution. The expected number of defective items can be zero defective, one defective, two defective items. Thus, a random variable may take values 0, 1 and 2.

$$\text{Now } P_1 = P(X=0) = \frac{C(6,3) \times C(2,0)}{C(8,3)}$$

$$= \frac{6!}{3! \times 2!} \times \frac{2!}{2! \times 0!} \times \frac{1}{8!} = \frac{20}{56}$$

$$P_2 = P(X=1) = \frac{C(6,2) \times C(2,1)}{C(8,3)}$$

$$= \frac{6!}{4! \times 1!} \times \frac{2!}{1! \times 1!} \times \frac{1}{8!} = \frac{30}{56}$$

$$P_3 = P(X=2) = \frac{C(6,1) \times C(2,2)}{C(8,3)} = \frac{6!}{5! \times 1!} \times \frac{2!}{2! \times 0!} \times \frac{1}{8!} = \frac{6}{56}$$

Hence, the expected number of defective items drawn is

$$E(X) = P_1 X_1 + P_2 X_2 + P_3 X_3$$

$$= \frac{20}{56} \times 0 + \frac{30}{56} \times 1 + \frac{6}{56} \times 2 = \frac{42}{56} = \frac{3}{4}$$

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Example 1.44. A player tossed two coins. If two heads show he wins ₹ 4. If one head shows he wins ₹ 2, but if two tails show he pays ₹ 3 as penalty. Calculate the expected value of the game to him.

Solution. Here X takes the values 0, 1, 2

$$\text{Also } P_1 = P(X = \text{zero head}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

$$P_2 = P(X = \text{one head}) = \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{1}{2}$$

$$P_3 = P(X = \text{two heads}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

Also when

$$X = 0 \Rightarrow x_1 = ₹ 3$$

$$X = 1 \Rightarrow x_2 = ₹ 2$$

$$X = 2 \Rightarrow x_3 = ₹ 4$$

We want to find out

$$E(X) = P_1 x_1 + P_2 x_2 + P_3 x_3 = \frac{1}{4} \times (-3) + \frac{1}{2} \times 2 + \frac{1}{4} \times 4 = \frac{5}{4} = 1.25$$

Hence,

$$E(X) = ₹ 1.25.$$

Example 1.45. Find the mean and variance of uniform probability distribution $f(x) = \frac{1}{n}$ for $x = 1, 2, \dots, n$.

Solution.

X	1	2	3	4	5	6	...	n
$P(X)$	$\frac{1}{n}$	$\frac{1}{n}$	$\frac{1}{n}$	$\frac{1}{n}$	$\frac{1}{n}$	$\frac{1}{n}$	...	$\frac{1}{n}$

$$\text{Mean} = E(X) = \frac{1}{n} + \frac{2}{n} + \frac{3}{n} + \dots + \frac{n}{n}$$

$$= \frac{1 + 2 + 3 + \dots + n}{n}$$

$$= \frac{1}{n} \left[\frac{n(n+1)}{2} \right] = \frac{n+1}{2}$$

$$\text{Variance} = E(X^2) - [E(X)]^2$$

$$= \frac{1^2 + 2^2 + \dots + n^2}{n} - \left[\frac{n+1}{2} \right]^2$$

$$= \frac{n(n+1)(2n+1)}{6n} - \left[\frac{n+1}{2} \right]^2$$

$$= \frac{n+1}{2} \left[\frac{2n+1}{3} - \frac{(n+1)}{2} \right] = \frac{n^2-1}{6}$$

Example 1.46. For discrete probability distribution.

x	0	1	2	3	4	5	6	7
f	0	k	2k	2k	3k	k ²	2k ²	7k ³

Determine: (i) k, (ii) mean, (iii) variance, (iv) smallest value of x such that $P(X \leq x) = 1$

Solution. (i)

$$\sum f(x) = 1$$

$$0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^3 = 1$$

$$10k^3 + 9k^2 - 1 = 0$$

$$k = -1 \text{ or } \frac{1}{10}$$

$\therefore k = 1$ since probability can never be negative.

x	0	1	2	3	4	5	6	7
f	0	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{1}{100}$	$\frac{2}{100}$	$\frac{7}{100}$

(ii) Mean = $\sum x f(x)$

$$= 0 + 1 \times \frac{1}{10} + 2 \times \frac{2}{10} + 3 \times \frac{2}{10} + 4 \times \frac{3}{10} + 5 \times \frac{1}{100} + 6 \times \frac{2}{100} + 7 \times \frac{7}{100}$$

$$= 3.66$$

(iii) Variance = $E(X^2) - [E(X)]^2$

$$= \left[0 + 1^2 \times \frac{1}{10} + 2^2 \times \frac{2}{10} + 3^2 \times \frac{2}{10} + 4^2 \times \frac{3}{10} + 5^2 \times \frac{1}{100} + 6^2 \times \frac{2}{100} + 7^2 \times \frac{7}{100} \right] - (3.66)^2$$

$$= 37.7$$

(iv)

$$P(X \leq 0) = f(0) = 0$$

$$P(X \leq 1) = f(0) + f(1) = 0.1$$

$$P(X \leq 2) = 0 + 0.1 + 0.2 = 0.3$$

$$P(X \leq 3) = 0.3 + 0.2 = 0.5$$

$$P(X \leq 4) = 0.5 + 0.3 = 0.8$$

$\therefore$ Smallest value of X such that $P(X \leq x) \geq 0.5$ is 4.

Example 1.47. Two dice are thrown. Let X assign to each point (a, b) in S the maximum number i.e., $X(a, b) = \max(a, b)$. Find the probability distribution of random variable X. $X(S) = \{1, 2, 3, 4, 5, 6\}$.

Solution. When two dice are thrown sample space S is

$$S = \left\{ \begin{array}{l} (1,1), (1,2), (1,3), (1,4), (1,5), (1,6) \\ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6) \\ (3,1), (3,2), (3,3), (3,4), (3,5), (3,6) \\ (4,1), (4,2), (4,3), (4,4), (4,5), (4,6) \\ (5,1), (5,2), (5,3), (5,4), (5,5), (5,6) \\ (6,1), (6,2), (6,3), (6,4), (6,5), (6,6) \end{array} \right\}$$

From the data $X(1, 1) = \max(1, 1) = 1$

$$P(X=1) = 1/36$$

$$X(1, 2), (2, 2), (1, 2) = 2, P(X=2) = 3/36$$

$$X(1, 3), (3, 3), (2, 3), (3, 2) = 3, P(X=3) = 5/36$$

$$X(1, 4), (4, 4), (3, 4), (4, 3), (2, 4), (4, 2) = 4, P(X=4) = 7/36$$

$$X(1, 5), (5, 5), (4, 5), (5, 4), (3, 5), (5, 3), (4, 4), (5, 4), (4, 5) = 5, P(X=5) = 9/36$$

$$X(1, 6), (6, 6), (5, 6), (6, 5), (4, 6), (6, 4), (5, 5), (6, 5), (5, 6) = 6, P(X=6) = 11/36$$

$\therefore$ Probability distribution to random variable X is

X	1	2	3	4	5	6
P(X)	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

Example 1.48. A fair coin is tossed until head or five tails occurs. Find expected number of tosses of the coins.

Solution. Probability of getting head = $1/2 = p$

Probability of getting tail = $1/2 = q$

X	1	2	3	4	5	6
Outcome	H	TH	TTH	TTTH	TTTTH	TTTTT
Probability	p	qp	q ² p	q ³ p	q ⁴ p	q ⁵

The expected number of tosses are

$$E(X) = \sum x_i p_i$$

$$= 1 \cdot p + 2qp + 3q^2p + 4q^3p + 5q^4p + 6q^5$$

$$= 1 \cdot \frac{1}{2} + 2 \left(\frac{1}{2} \right)^2 + 3 \left(\frac{1}{2} \right)^3 + 4 \left(\frac{1}{2} \right)^4 + 5 \left(\frac{1}{2} \right)^5 + 6 \left(\frac{1}{2} \right)^5$$

$$= 1.9687 = 2$$

Therefore expected number of tosses are 2.

Example 1.49. Two cards are drawn successively with replacement from a well shuffled pack of 52 cards. Find the mean and variance of the number of kings.

Solution. Let X be the random variable. The

X = Number of kings obtained in two draws

Clearly, X can assume the value 0, 1, 2

$$P(\text{drawing a king}) = \frac{4}{52} = \frac{1}{13}$$

$$P(\text{not drawing a king}) = 1 - \frac{1}{13} = \frac{12}{13}$$

$$P(X=0) = P(\text{not a king in the 1st draw and not a king in 2nd draw})$$

$$= \left(\frac{12}{13} \times \frac{12}{13} \right) = \frac{144}{169}$$

$P(X=1) = P(\text{a king in the 1st draw and not a king in the 2nd draw})$
 or $P(\text{not a king in the 1st draw and a king in the 2nd draw})$

$$= \left(\frac{1}{13} \times \frac{12}{13} + \frac{12}{13} \times \frac{1}{13} \right) = \frac{24}{169}$$

$$P(X=2) = P(\text{a king in the 1st draw and a king in the 2nd draw})$$

$$= \left(\frac{1}{13} \times \frac{1}{13} \right) = \frac{1}{169}$$

Hence, the probability distribution is given by

$X - Y_i$	0	1	2
P_i	$\frac{144}{169}$	$\frac{24}{169}$	$\frac{1}{169}$

$\therefore$ Mean,

$$\mu = \sum P_i Y_i = \left(\frac{144}{169} \times 0 \right) + \left(\frac{24}{169} \times 1 \right) + \left(\frac{1}{169} \times 2 \right) = \frac{2}{13}$$

Variance,

$$\sigma^2 = \sum P_i Y_i^2 - \mu^2$$

$$= \left[\left(\frac{144}{169} \times 0 \right) + \left(\frac{24}{169} \times 1 \right) + \left(\frac{1}{169} \times 4 \right) \right] - \frac{4}{169} = \frac{24}{169}$$

Example 1.50. An industrial salesman wants to know the average number of units he sells per sales call. He checks his past sales records and comes up with the following probability

Sales in units	0	1	2	3	4	5
Probability	0.15	0.20	0.10	0.05	0.30	0.20

What is the average number of units he sells per sales call?

Solution. The sales wants to know the average number of units he sells per sales call. This is same thing as saying that he wants to know the expected value of each sales call, where a sales call is the random variable X . The expected value is calculated by the formula :

$$E(X) = P_1 X_1 + P_2 X_2 + P_3 X_3 + \dots$$

$$= 0.15 \times 0 + 0.20 \times 1 + 0.10 \times 2 + 0.05 \times 3 + 0.30 \times 4 + 0.20 \times 5$$

$$= 0 + 0.2 + 0.2 + 0.15 + 1.0 = 2.75$$

Thus he would expect to sell 2.75 units on each sales call.

Example 1.51. A die is tossed twice. A 'success' is getting an odd number on a random toss. Find the variance of the number of successes.

Solution.

$$P(\text{success}) = \frac{3}{6} = \frac{1}{2}$$

and

$$P(\text{failure}) = 1 - \frac{1}{2} = \frac{1}{2}$$

$\therefore$

$$P(X=0) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

$$P(X=1) = \frac{0}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{1}{2}$$

$$P(X=2) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

Hence, the probability distribution is given by

X	0	1	2
$P(X)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

$\therefore$

$$\text{Mean, } \mu = \sum P_i Y_i = \frac{1}{4} \times 0 + \frac{1}{2} \times 1 + \frac{1}{4} \times 2 = 1$$

Variance,

$$\sigma^2 = \sum P_i Y_i^2 - \mu^2 = \left[\left(\frac{1}{4} \times 0 \right) + \left(\frac{1}{2} \times 1 \right) + \left(\frac{1}{4} \times 4 \right) \right] - 1^2$$

$$= \frac{3}{4} - 1 = -\frac{1}{4}$$

Example 1.52. Find the expected value of the absolute difference of upturned faces in the experiment of tossing of two dice.

Solution.

X	Abs. Diff.	Total no. of cases	$P(X)$
0	(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)	6	$6/36$
1	(1, 2), (2, 3), (3, 4), (4, 5), (5, 6) (2, 1), (3, 2), (4, 3), (5, 4), (6, 5)	10	$10/36$
2	(1, 3), (2, 4), (3, 5), (4, 6), (3, 1), (4, 2), (5, 3), (6, 4)	8	$8/36$
3	(1, 4), (2, 5), (3, 6), (4, 1), (5, 2), (6, 3)	6	$6/36$
4	(1, 5), (2, 6), (5, 1), (6, 2)	4	$4/36$
5	(1, 6), (6, 1)	2	$2/36$

Hence,

$$E(X) = \sum_{i=1}^n x_i P_i(X)$$

$$= 0 \times \frac{6}{36} + 1 \times \frac{10}{36} + 2 \times \frac{8}{36} + 3 \times \frac{6}{36} + 4 \times \frac{4}{36} + 5 \times \frac{2}{36}$$

$$= \frac{70}{36} = \frac{35}{18}$$

Example 1.53. If X and Y are discrete random variables and K is a constant then prove that
 (i) $E(X + K) = E(X) + K$ and (ii) $E(X + Y) = E(X) + E(Y)$.

Solution. Given X and Y are discrete random variable and K is a constant.

Since

$$E(X) = \sum_{i=1}^n x_i P_i(X) \text{ and } \sum_{i=1}^n P_i = 1$$

$$E(X+K) = \sum_{i=1}^n (X_i + K)P_i$$

$$= \sum_{i=1}^n X_i P_i + K \sum_{i=1}^n P_i = E(X) + K \cdot 1$$

$$\therefore \sum_{i=1}^n P_i = 1$$

$$E(X+K) = E(X) + K$$

$$(X+Y) = \sum_{i=1}^n (x_i + y_i)P_i = \sum_{i=1}^n x_i P_i + \sum_{i=1}^n y_i P_i$$

$$= E(X) + E(Y)$$

$$E(X+Y) = E(X) + E(Y)$$

EXERCISE 1.3

1. A coin is tossed four times. If X is the number of heads observed, find the probability distribution of X .

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ \text{Ans. } \frac{1}{16} & \frac{1}{4} & \frac{3}{8} & \frac{1}{4} & \frac{1}{16} \end{bmatrix}$$

2. An urn contains 5 red and 2 black balls. Two balls are randomly selected. Let X represent the number of black balls. What are the possible values of X ? Is X a random variable?

$$\begin{bmatrix} \text{Ans. } \frac{10}{21} & \frac{10}{21} & \frac{1}{21} \end{bmatrix}$$

3. A random variable x has the following probability distribution.

x	0	1	2	3	4	5	6	7
$P(x)$	a	$4a$	$3a$	$7a$	$8a$	$10a$	$6a$	$9a$

(i) Determine the value of a .

(ii) Find $P(X < 3)$, $P(X \geq 4)$, $P(0 < X < 5)$

(iii) Give the smallest value of m for which $P(X \leq m) \leq 0.6$.

4. A person plays a game of tossing a coin thrice. For each head he gets ₹ 2 from the organiser, and for each tail he has to give ₹ 1.50 to the organiser. Let X denote the amount gained or lost by the person. Show that X is a random variable and exhibit it as a function on the sample space.

5. Find the probability distribution to the number of doubles in three throws of a pair of dice.

$$\begin{bmatrix} 0 & 1 & 2 & 3 \\ \text{Ans. } \frac{125}{216} & \frac{75}{216} & \frac{15}{216} & \frac{1}{216} \end{bmatrix}$$

BASIC PROBABILITY

6. Find the probability distribution of the number of successes in two tosses of a die when a success is defined as getting a value 5 or 6.

$$\begin{bmatrix} 0 & 1 & 2 \\ \text{Ans. } \frac{4}{9} & \frac{4}{9} & \frac{1}{9} \end{bmatrix}$$

7. Two cards are drawn successively with replacement from a well-shuffled deck of 52 cards. Find the probability distribution of the number of jacks.

$$\begin{bmatrix} 0 & 1 & 2 \\ \text{Ans. } \frac{144}{169} & \frac{24}{169} & \frac{1}{169} \end{bmatrix}$$

8. A random variable X has the following probability distribution where k is some number.

$$P(x) = \begin{cases} k & \text{if } x = 0 \\ 2k & \text{if } x = 1 \\ 3k & \text{if } x = 2 \\ 0 & \text{otherwise} \end{cases}$$

(a) Determine the value of k .

$$\begin{bmatrix} \text{Ans. } k = \frac{1}{6} \\ \text{Ans. } \frac{1}{2}, \frac{1}{2} \end{bmatrix}$$

(b) Find $P(X < 2)$, $P(X \leq 2)$, $P(X \geq 2)$.

9. A box contains 13 bulbs out of which 5 bulbs are defective. 3 bulbs are drawn one by one from the box without replacement. Find the probability distribution of the number of defective bulbs drawn.

$$\begin{bmatrix} 0 & 1 & 2 & 3 \\ \text{Ans. } \frac{28}{143} & \frac{70}{143} & \frac{40}{143} & \frac{5}{143} \end{bmatrix}$$

10. Two bad eggs are accidentally mixed with ten good ones. Three eggs are drawn at random with replacement from this lot. Compute μ and σ^2 for the number of bad eggs drawn.

$$\begin{bmatrix} \text{Ans. } \frac{1}{2}, \frac{5}{12} \end{bmatrix}$$

11. A die is tossed thrice. A success is 'getting 1 or 6' on toss. Find the mean and the variance of the number of successes.

$$\begin{bmatrix} \text{Ans. } \mu = 1, \sigma^2 = \frac{2}{3} \end{bmatrix}$$

12. Compute the variance of the probability distribution of the number of doubles in four throws of a pair of dice.

x	1	2	3	4	8	9
$P(x)$	k	$3k$	$5k$	$7k$	$9k$	$11k$

$$\begin{bmatrix} \text{Ans. (i) } \frac{1}{36} \end{bmatrix}$$

Determine: (i) k , (ii) mean, (iii) $P(X) \geq 3$.

42. If 3 cars are drawn from a lot of 6 cars containing 2 defective cars, find :

14. If 3 cars are drawn from a lot of 6 cars containing 2 defective cars, find :
- (i) the probability distribution of the number of defective cars

	X	0	1	2
Ans. (i)	$P(X)$	$\frac{1}{5}$	$\frac{3}{5}$	$\frac{1}{5}$
				(iii)

- (ii) the expected number of defective cars.

15. A random variable X has the following distribution

x	0	1	2	3	4	5	6
$P(X)$	k	$3k$	$5k$	$7k$	$9k$	$11k$	$13k$

Find: (i) $P(X < 4)$, $P(X \geq 5)$, $P(3 < X \leq 6)$ and (ii) What will be the minimum value of X so that $P(X \geq 2) > 0.3$.

[Ans. (i) 0.3265, 0.4898, 0.6734; (ii) 3]

16. A player tosses two fair coins. He wins ₹ 100/- if head appears. ₹ 200/- if two heads appear. On the other hand he loses ₹ 500/- if no head appears. Determine the expected value E of the game and is the game favourable to the player?

[Ans. 25; Not favourable to the player]

1.12 INDEPENDENT RANDOM VARIABLE

Two random variable are said to be independent if the probability of either variable taking a particular value does not depend on the value taken by the other variable.

Theorem: The expectation of the product of two independent random variable is equal to product of their expectation.

i.e., $E(xy) = E(x) E(y)$ where x and y are independent random variables.

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BINOMIAL DISTRIBUTION

in each loss, $\frac{1}{2}$ of the loss

be called a success and $y = 0$ if the trial is a failure. If the trials are independent and the probability of success

$$C(n, r) p^r q^{n-r}$$
$$P(X=r) = P(r) = C(n,r)P^r q^{n-r}; r = 0, 1, 2, 3, 4, \dots, n,$$
 $(q + p)^2$

The Mineral distributions is given by

$$I(r) = NI^*(r) = NC^*(n, r) p^r q^{n-r}$$

Frequency distribution

1.18. PROPERTIES OF A BINOMIAL DISTRIBUTION

- 1 It is a discrete distribution which gives the theoretical probabilities.
- 2 It depends on the parameters p or q , the probability of success or failure and n (the number of trials). The parameter n is always a positive integer.
- 3 The distribution will be symmetrical if $p = q$.
- 4 The statistics of the binomial distribution are mean $= np$, variance $= npq$, and standard deviation $= \sqrt{npq}$.

As

BASIC PROBABILITY

- changes for a given
- p
- .

1.19. MEAN OF BINOMIAL DISTRIBUTION

$$P(X = r) = {}^nC_r p^r q^{n-r}$$

The discrete probability distribution for the binomial distribution can be displayed as follows:

X	0	1	2	...	r	...	n
$P(X)$	nC_0q^n	nC_1pq^{n-1}	$nC_2p^2q^{n-2}$	...	$nC_r p^r q^{n-r}$	...	$nC_n p^n$

$$\text{Mean } (\mu) = E(X) = \sum rP(X=r)$$

$$= {}^nC_0 q^n \times 0 + {}^nC_1 p q^{n-1} \times 1 + {}^nC_2 p^2 q^{n-2} \times 2 + \dots + {}^nC_r p^r q^{n-r} \times r + \dots + {}^nC_n p^n \times n$$

$$= 0 + npq^{n-1} + \frac{n(n-1)}{2!} p^2 q^{n-2} \times 2 + \dots + np^n$$

$$= np \left[q^{n-1} + (n-1)pq^{n-2} + \frac{(n-1)(n-2)}{2} p^2 q^{n-3} + \dots + p^{n-1} \right]$$

$$(u + b \therefore) du = \frac{1}{1-a} (d + b) du =$$

$$\therefore \text{Mean} = np$$

1.20. VARIANCE OF BINOMIAL DISTRIBUTION

Since $\text{Variance} = \Sigma p x^2 - \mu^2$

$$\Sigma p x^2 = {}^nC_0 q^n \times (0)^2 + {}^nC_1 p q^{n-1} \times (1)^2 + {}^nC_2 p^2 q^{n-2} \times (2)^2 + \dots$$

Now

$$+ nC_3 p^3 q^{n-3} \times (3)^2 + \dots + nC_n p^n$$

Breaking second, third and following terms into parts, we get

$$\Sigma p^{n^2} = np \left[q^{n-1} + (n-1)pq^{n-2} + \frac{(n-1)(n-2)}{2!} p^2 q^{n-3} + \dots + p^{n-1} \right] + \\ n(n-1)p^2 \left[q^{n-2} + (n-2)pq^{n-3} + \frac{(n-2)(n-3)}{2!} p^2 q^{n-4} + \dots \right]$$

$$=np(q+n)^{n-1}+n(n-1)p^2(q+p)^{n-2} \\ =np+n(n-1)p^2-np[1+(n-1)p] =np[q+np]npq+n^2p^2$$

$$\text{Variance} = n\sigma q + n^2 p^2 = (np)^2 = n\sigma q$$

1.21. MODE OF BINOMIAL DISTRIBUTION
Mode is the value of r at which $p(r)$ has maximum value. Let x be the mode of binomial distribution.

then $p(r) \geq p(r+1)$ and $p(r) \geq p(r-1)$

$$\Rightarrow \frac{p(r)}{p(r+1)} \geq 1$$

Consider $p(r) \geq p(r+1)$

$$\Rightarrow \frac{(r+1)q}{(n-r)p} \geq 1$$

$$\Rightarrow \frac{{}^nC_r p^r q^{n-r}}{{}^nC_{r+1} p^{r+1} q^{n-r-1}} \geq 1$$

$$\Rightarrow (r+1)q \geq (n-r)p$$

$$\Rightarrow r \geq np - 1 + p$$

$\Rightarrow$

$$p(r) \geq p(r-1) \Rightarrow \frac{p(r)}{p(r-1)} \geq 1$$

$$\Rightarrow \frac{{}^nC_r p^r q^{n-r}}{{}^nC_{r-1} p^{r-1} q^{n-r+1}} \geq 1$$

$$\Rightarrow \frac{n-r+1}{r} \cdot \frac{p}{q} \geq 1$$

$$\Rightarrow (n-r+1)p \geq qr$$

$$\Rightarrow r \leq (n+1)p$$

From equations (i) and (ii)

$$\{(n+1)p - 1\} \leq r \leq (n+1)p$$

Case (i): If $(n+1)p$ is not an integer, then mode is the integral part of $(n+1)p$. In this case the distribution is called 'unimodal'.

Case (ii): If $(n+1)p$ is an integer then both $(n+1)p$ and $\{(n+1)p - 1\}$ will represent modes. In this case the distribution is called 'bimodal'.

Constants of Binomial Distribution

$$\text{Mean} = np$$

$$\text{Standard deviation} = \sqrt{npq}$$

$$\text{First moment or } \mu_1 = 0$$

$$\text{Second moment or } \mu_2 = npq$$

$$\text{Third moment or } \mu_3 = npq(q-p)$$

$$\text{Fourth moment or } \mu_4 = 3n^2 p^2 + npq(1-6pq)$$

$$\beta_1 = \frac{(q-p)^2}{npq}$$

$$\beta_2 = 3 + \frac{1-6pq}{npq}$$

1.22. CONDITIONS FOR APPLICATION OF BINOMIAL DISTRIBUTION

1. The variable should be discrete i.e., defectives should be 1, 2, 3, 4 or 5 etc., and never 1.5, 2.1 or 3.41 etc.
2. A dichotomy exists. In other words, the happening of events must be of two alternative. It must be either a success or failure.
3. The number of trials n should be finite and small.
4. The trials or events must be independent. The happening of one event must not affect the happening of other events. In other words, statistical independence must exist.
5. The trial or events must be repeated under identical conditions.

1.23. RECURSION FORMULA OR RECURRENCE RELATION FOR BINOMIAL DISTRIBUTION

We know that for the Binomial distribution

$$P(X=r) = {}^nC_r p^r q^{n-r}$$

$$\text{and } P(X=r+1) = {}^nC_{r+1} p^{r+1} q^{n-r-1}$$

$$\Rightarrow \frac{P(X=r+1)}{P(X=r)} = \frac{{}^nC_{r+1} p^{r+1} q^{n-r-1}}{{}^nC_r p^r q^{n-r}}$$

$$= \frac{n!}{(r+1)!(n-r-1)!} \times \frac{r!(n-r)!}{n!} \times \frac{p^{r+1} q^{n-r-1}}{p^r q^{n-r}} = \frac{n-r}{r+1} \cdot \frac{p}{q}$$

$$\Rightarrow P(X=r+1) = \frac{n-r}{r+1} \cdot \frac{p}{q} P(X=r); r=1, 2, 3, \dots$$

which is the required recurrence formula. Applying this formula successively, we can find $P(X=1)$, $P(X=2)$, $P(X=3)$... if $P(X=0)$ is known.

SOLVED EXAMPLES

Example 1.54. Ten coins are thrown simultaneously. Find the probability of getting at least seven heads.

Solution. When one coin is thrown,

$$\text{The probability of getting a head} = \frac{1}{2}$$

$$\therefore \text{The probability of getting a tail} = \frac{1}{2}$$

$$\text{The probability of not getting a head} = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\therefore \text{The probability of getting a head} = \frac{1}{2}$$

Then P (at least 7 heads) = P (7 heads) + P (8 heads) + P (9 heads) + P (10 heads)

$$= {}^{10}C_7 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^3 + {}^{10}C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^2 + {}^{10}C_9 \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right)^1 + {}^{10}C_{10} \left(\frac{1}{2}\right)^{10}$$

$$= \frac{1}{2^{10}} [{}^{10}C_7 + {}^{10}C_8 + {}^{10}C_9 + {}^{10}C_{10}]$$

$$= \frac{120 + 45 + 10 + 1}{1024} = \frac{176}{1024} = \frac{11}{64}$$

Example 1.55. In a lot of 200 articles 10 are defective, find the probability of: (i) no defective article, (ii) one defective article, (iii) at least one defective article, in a random sample of 20 articles.

$$\frac{10}{200} = \frac{1}{20}$$

Solution. The probability of defective article is $\frac{1}{20}$

$$p = \frac{1}{20}$$

$$\therefore 1 - \frac{1}{20} = \frac{19}{20} \Rightarrow q = \frac{19}{20}$$

The probability of non-defective article = $1 - \frac{1}{20} = \frac{19}{20}$

$$(i) \text{ The probability of no defective article out of 20}$$

$$= {}^{20}C_0 (p)^0 (q)^{20} = \left(\frac{19}{20}\right)^{20}$$

[$\because p_0 q_1$]

$$(ii) \text{ The probability of exactly one defective article}$$

$$= {}^{20}C_1 (p)^1 (q)^{19} = 20 \times \frac{1}{20} \times \left(\frac{19}{20}\right)^{19} = \left(\frac{19}{20}\right)^{19}$$

$$(iii) \text{ The probability of at least one will be defective}$$

$$= 1 - [\text{probability that none will be defective}]$$

$$= 1 - {}^{20}C_{20} \left(\frac{19}{20}\right)^{20} = 1 - \left(\frac{19}{20}\right)^{20}$$

Example 1.56. If on an average, one ship out of 10 is wrecked, find the probability that out of 5 ships expected to arrive the port, at least four will arrive safely.

Solution. p be the probability of a ship arriving safely = $1 - \frac{1}{10} = \frac{9}{10}$

$$q = 1 - \frac{9}{10} = \frac{1}{10}$$

$$\text{Binomial distribution is } \left(\frac{1}{10} + \frac{9}{10}\right)^5$$

Probability that at least four ships out of five arrive safely

$$= P(4) + P(5) = {}^5C_4 \left(\frac{1}{10}\right)^1 \left(\frac{9}{10}\right)^4 + {}^5C_5 \left(\frac{1}{10}\right)^0 \left(\frac{9}{10}\right)^5$$

$$= \left(\frac{9}{10}\right)^4 \frac{14}{10} = \left(\frac{9}{10}\right)^4 \frac{7}{5} = 0.91854$$

Example 1.57. The probability that a man aged 60 will live to the 70 is 0.65. What is the probability that out of 10 men aged 60 now, at least 7 would live to the 70?

Solution. Probability of survival upto the age of 70

$$= p = 0.65$$

Probability of non-survival upto the age of 70

$$= q = 1 - p = 1 - 0.65 = 0.35$$

Probability that out of 10 such men at least 7 would survive as desired

$$= \text{Probability that exactly 7 would survive} +$$

$$+ \text{Probability that exactly 8 would survive} +$$

$$+ \text{Probability that exactly 9 would survive} +$$

$$+ \text{Probability that exactly 10 would survive}$$

$$= P(7) + P(8) + P(9) + P(10)$$

$$= {}^{10}C_7 p^7 q^3 + {}^{10}C_8 p^8 q^2 + {}^{10}C_9 p^9 q + {}^{10}C_{10} p^{10}$$

$$= 120 p^7 q^3 + 45 p^8 q^2 + 10 p^9 q + p^{10}$$

$$= p^7 (120 q^3 + 45 p q^2 + 10 p^2 q + p^3)$$

$$= (0.65)^7 [120 \times (0.35)^3 + 45 (0.65) (0.35)^2$$

$$+ 10 (0.65)^2 (0.35) + (0.65)^3]$$

$$= 0.514, \text{ the required result.}$$

Example 1.58. Six dice are thrown together at a time, the process is repeated 729 times. How many times do you expect at least three dice to have 4 to 6?

Solution. The chance of getting 4 or 6 with one dice is

$$\frac{2}{6} \text{ i.e., } p = \frac{1}{3} \text{ and } q = 1 - \frac{1}{3} = \frac{2}{3}$$

In one throw of six dice together, we have probability of getting at least 3 dice to have 4 or 6.

$$= P(3) + P(4) + P(5) + P(6)$$

$$= {}^6C_3 p^3 q^3 + {}^6C_4 p^4 q^2 + {}^6C_5 p^5 q + {}^6C_6 p^6$$

$$= 20 \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^3 + 15 \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^2 + 6 \left(\frac{1}{3}\right)^5 \left(\frac{2}{3}\right) + \left(\frac{1}{3}\right)^6$$

$$= \frac{1}{(3)^6} [160 + 60 + 12 + 1] = \frac{233}{(3)^6}$$

Now the process is repeated 729 times

$\therefore$ Required number of times at least 3 dice have 4 or 6

$$= 729 \times \frac{233}{(3)^6} = 233, \text{ the required result.}$$

Note. In the above case the binomial distribution is $N(q + p)^n$ where $N = 729$, $n = 6$

Example 1.59. If the sum of the mean and the variance of binomial distribution of 5 trials is 4.8, find the distribution.

Solution. Let the required binomial distribution be ${}^nC_r p^r q^{n-r}$ where n = number of trials = 5

Mean of the distribution = np
and the variance of the distribution = npq

By the given condition

$$np + npq = 4.8$$

$$5p + 5pq = 4.8$$

$$5p(1+q) = 4.8$$

$$50(1-q^2) = 48 \Rightarrow 50 - 50q^2 = 48$$

$$50q^2 = 2 \Rightarrow q = \frac{1}{5}$$

$$p = 1 - q = 1 - \frac{1}{5} = \frac{4}{5}$$

Hence, the required binomial distribution is ${}^5C_r \left(\frac{4}{5}\right)^r \left(\frac{1}{5}\right)^{5-r}$

Example 1.60. The probability that a bomb dropped from a plane will strike the target is $\frac{1}{5}$. If six bombs are dropped, find the probability that: (i) exactly two will strike the target, (ii) at least two will strike the target.

Solution. The probabilities of 0, 1, 2 ... successes are given by the respective terms in the expansion of

$$(q+p)^n = \left(\frac{4}{5} + \frac{1}{5}\right)^6, \text{ since } p = \frac{1}{5}, q = \frac{4}{5} \text{ and } n = 6.$$

$\therefore P(2)$ = The probability that exactly two bombs will strike the target

$$= {}^6C_2 p^2 q^4 = \frac{6 \cdot 5}{1 \cdot 2} \left(\frac{1}{5}\right)^2 \left(\frac{4}{5}\right)^4 = 0.246$$

The probability that at least 2 bombs will strike the target

$$= 1 - [P(0) + P(1)]$$

$$= 1 - q^6 - {}^6C_1 q^5 p = 1 - (0.8)^6 - 6(0.2)(0.8)^5$$

$$= 1 - 0.2621 - 0.3932 = 0.3447$$

Example 1.61. Assuming that half the population are consumers of rice so that the chance of an individual being a rice consumer is $\frac{1}{2}$ and assuming that 100 investigations each take 11 individuals to see whether they are rice consumers, how many investigations would you expect to report that three people or less consumers?

Solution. Here $p = \frac{1}{2}, q = \frac{1}{2}, n = 10, N = 100$

$\therefore$ The probability that r persons out of 10 persons are consumers of rice is given by

$$P(r) = {}^{10}C_r \left(\frac{1}{2}\right)^r \left(\frac{1}{2}\right)^{10-r}$$

$\therefore$ The expected number of investigators (i.e., expected frequencies) who would report that three or less people were consumers of rice.

$$= 100 [P(0) + P(1) + P(2) + P(3)]$$

$$= 100 \left[{}^{10}C_0 \left(\frac{1}{2}\right)^0 + {}^{10}C_1 \left(\frac{1}{2}\right)^1 + {}^{10}C_2 \left(\frac{1}{2}\right)^2 + {}^{10}C_3 \left(\frac{1}{2}\right)^3 \right]$$

$$= \frac{100}{2^{10}} [1 + 10 + 45 + 120] = \frac{17600}{1024} = 17 \text{ approx.}$$

Example 1.62. A die is thrown 5 times. Getting an even number greater than 2 is considered a success. Calculate $P(X=r)$ for $r = 1, 2, 3, 4, 5$ from recurrence formula.

Solution. Let p be the probability of getting an even number greater than 2 on a die.

$$\Rightarrow p = \frac{2}{6} = \frac{1}{3}$$

$$q = 1 - p \Rightarrow q = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\therefore \frac{p}{q} = \frac{1}{2}. \text{ Also } n = 5$$

$P(X) = 0$ = Probability of no success in 5 trials

$$= {}^5C_0 (q)^5 = \left(\frac{2}{3}\right)^5 = 0.1317$$

Recurrence formula for binomial distribution is

$$P(X=r+1) = \frac{n-r}{r+1} \cdot \frac{p}{q} P(X=r)$$

$$= \frac{5-r}{r+1} \cdot \left(\frac{1}{2}\right) P(X=r)$$

$$\text{Putting } r = 0, \text{ in (i), } P(X=1) = 5 \left(\frac{1}{2}\right) P(X=0) = 5 \left(\frac{1}{2}\right) (0.1317) = 0.3292$$

$$\text{Putting } r = 1, \text{ in (i), } P(X=2) = 2 \left(\frac{1}{2}\right) P(X=1) = P(X=1) = 0.3292$$

$$\text{Putting } r = 2, \text{ in (i), } P(X=3) = (1) \left(\frac{1}{2}\right) P(X=2) = \frac{1}{2} \cdot (0.3292) = 0.1646$$

$$\text{Putting } r = 3, \text{ in (i), } P(X=4) = \frac{2}{4} \cdot \frac{1}{2} P(X=3) = \frac{1}{4} (0.1646) = 0.0412$$

$$\text{Putting } r = 4, \text{ in (i), } P(X=5) = \frac{1}{5} \cdot \frac{1}{2} P(X=4) = \frac{1}{10} (0.0412) = 0.0041.$$

Example 1.63. Out of 800 families with 4 children each, how many families would be expected to have: (i) 2 boys and 2 girls, (ii) at least one boy, (iii) no girl, (iv) at most two girls? Assume the probabilities for boys and girls are equal.

Solution. Since probability for boys and girls are equal

$$p = \text{Probability of having a boy} = \frac{1}{2}$$

$$q = \text{Probability of having a girl} = \frac{1}{2}$$

$$n = 4, N = 800$$

The binomial distribution is $800 \left(\frac{1}{2} + \frac{1}{2} \right)^4$.

(i) The expected number of families having 2 boys and 2 girls

$$= 800 \cdot {}^4C_2 \left(\frac{1}{2} \right)^2 \left(\frac{1}{2} \right)^2 = 800 \times 6 \times \frac{1}{16} = 300$$

(ii) The expected number of families having at least one boy

$$= 800 \left[{}^4C_1 \left(\frac{1}{2} \right)^3 \left(\frac{1}{2} \right) + {}^4C_2 \left(\frac{1}{2} \right)^2 \left(\frac{1}{2} \right)^2 + {}^4C_3 \left(\frac{1}{2} \right) \left(\frac{1}{2} \right)^3 + {}^4C_4 \left(\frac{1}{2} \right)^4 \right]$$

$$= 800 \times \frac{1}{16} [4 + 6 + 4 + 1] = 750$$

(iii) The expected number of families having no girl having 4 boys

$$= 800 \times {}^4C_4 \left(\frac{1}{2} \right)^4 = 50$$

(iv) The expected number of families having at most two i.e., having at least 2 boys

$$= 800 \left[{}^4C_2 \left(\frac{1}{2} \right)^2 \left(\frac{1}{2} \right)^2 + {}^4C_3 \left(\frac{1}{2} \right) \left(\frac{1}{2} \right)^3 + {}^4C_4 \left(\frac{1}{2} \right)^4 \right]$$

$$= 800 \times \frac{1}{16} [6 + 4 + 1] = 550.$$

Example 1.64. A student obtained the following answer to a certain problem given to him
Mean = 2.4; variance = 3.2 for a binomial distribution. Comment on the result.

Solution. The mean of binomial distribution is np and variance npq . We are given mean $= np = 2.4$

$$\text{Variance} = npq$$

$$2.4q = 3.2$$

$$q = \frac{3.2}{2.4} = 1.333$$

Since the value of q is greater than 1, the given results are inconsistent.

Example 1.65. Ten coins are tossed 1024 times and the following frequencies are observed. Compare these frequencies with the expected frequencies:

Number of heads	0	1	2	3	4	5	6	7	8	9	10
Frequencies	2	10	38	106	188	257	226	128	59	7	3

Solution. Here $n = 10, N = 1024$

$$p = \text{The change of getting a head in one toss} = \frac{1}{2}$$

$$q = 1 - p = \frac{1}{2}$$

The expected frequencies are the respective terms of the binomial $1024 \left(\frac{1}{2} + \frac{1}{2} \right)^{10}$

The frequency of r heads ($0 \leq r \leq 10$) is

$$= 1024 \cdot {}^{10}C_r \left(\frac{1}{2} \right)^{10-r} \left(\frac{1}{2} \right)^r = 1024 \times {}^{10}C_r \left(\frac{1}{2} \right)^{10} = {}^{10}C_r$$

Hence, we have the following comparison.

Number of heads	0	1	2	3	4	5	6	7	8	9	10
Observed frequency	2	10	38	106	188	257	226	128	59	7	3
Expected frequency	1	10	45	120	210	252	210	120	45	10	1

(${}^{10}C_0, {}^{10}C_1, {}^{10}C_2$ and so on.)

Example 1.66. Probability of man hitting a target is $\frac{1}{3}$.

(a) If the fires of 6 times, what is the probability of hitting: (i) at most 5 times, (ii) at least times, (iii) exactly once

(b) If he fires so that the probability of his hitting target atleast once is greater than $\frac{3}{4}$, find

Solution. (a) Given $p = 1/3, q = 1 - 1/3 = 2/3, n = 6$

(i) The probability of hitting the target almost 5 times.

$$P(X \leq 5) = 1 - P(X > 5) = 1 - P(X = 6)$$

$$= 1 - \left[{}^6C_6 \left(\frac{1}{3} \right)^6 \left(\frac{2}{3} \right)^0 \right] = 1 - \frac{1}{729} = \frac{728}{729}$$

(ii) The probability of hitting the target atleast 5 times

$$P(X \geq 5) = P(X = 5) + P(X = 6)$$

$$= {}^6C_5 \left(\frac{1}{3} \right)^5 \left(\frac{2}{3} \right)^1 + {}^6C_6 \left(\frac{1}{3} \right)^6 \left(\frac{2}{3} \right)^0 = \frac{13}{729}$$

(iii) The probability of hitting the target exactly once

$$P(X=1) = {}^nC_1 \left(\frac{1}{3}\right) \left(\frac{2}{3}\right)^4 = \frac{192}{729}$$

(iv) If the fires so that the probability of his hitting the target atleast once is greater than $3/4$ then

$$P(X \geq 1) = 3/4$$

$$\Rightarrow 1 - P(X < 1) > 3/4$$

$$\Rightarrow 1 - P(X = 0) > 3/4$$

$$\Rightarrow 1 - (2/3)^n > 3/4$$

$$\Rightarrow 1/4 > (2/3)^n$$

$$\Rightarrow \frac{1}{2^2} > \left(\frac{2}{3}\right)^n$$

$$\Rightarrow 3n > 2^{n+2}$$

$\Rightarrow$ This inequality is satisfied for $n = 4$

$\therefore$ He must fire 4 times to that probability of hitting the target atleast once is greater than $3/4$.

Example 1.67. A student takes a true-false examination consisting of 8 questions. He guesses each answer. The guesses are made at random. Find the smallest value of n so that the probability of guessing atleast n correct answers is less than $1/2$.

Solution. For given data, we have to find

$$P(X \geq n) < 1/2$$

Now probability of guessing a correct answer is $1/2$ and guessing a wrong answer is $1/2$

$$p = 1/2, q = 1/2, n = 8$$

Using Binomial distribution.

$$P(X \geq n) < 1/2$$

$$1 - P(X < n) < 1/2$$

$$\Rightarrow 1 - [P(X = 0) + P(X = 1) + \dots + P(X = n-1)] < 1/2$$

$$\Rightarrow P(X = 0) + P(X = 1) + \dots + P(X = n-1) > 1/2$$

$$\Rightarrow {}^nC_0 \left(\frac{1}{2}\right)^0 + {}^nC_1 \left(\frac{1}{2}\right)^1 + \dots + {}^nC_{n-1} \left(\frac{1}{2}\right)^{n-1} > \frac{1}{2}$$

$$\Rightarrow \left(\frac{1}{2}\right)^n [{}^nC_0 + {}^nC_1 + \dots + {}^nC_{n-1}] > \frac{1}{2}$$

This inequality is satisfied in $n-1 = 4$

$$n = 5$$

Example 1.68. Fit a binomial distribution to following data, when tossing 5 coins.

x	0	1	2	3	4	5
f	2	14	20	34	22	8

BASIC PROBABILITY

Solution.

$$\Sigma f = 100 = N$$

$$n = 5$$

$$\bar{x} = \text{mean}$$

$$= \frac{\Sigma fx}{\Sigma f} = \frac{0 \times 2 + 1 \times 14 + \dots + 5 \times 8}{100}$$

$$= 2.84$$

$$\text{Mean} = \bar{x} = np = 2.84$$

$$p = 0.57 \text{ and } q = 0.43$$

$\Rightarrow$ Using binomial distribution

$$P(X = x) = {}^nC_x p^x q^{n-x}; x = 0, 1, 2, \dots, n$$

and expected frequency obtained from

$$f(x) = N \cdot P(x)$$

x	f	$P(x) = {}^nC_x p^x q^{n-x}$	$f(x) = N \cdot P(x)$
0	2	${}^5C_0 (0.57)^0 (0.43)^5 = (0.43)^5$	$100 \times (0.43)^5 \approx 1$
1	14	${}^5C_1 (0.57)^1 (0.43)^4 = 0.098$	$100 \times (0.098) \approx 10$
2	20	${}^5C_2 (0.57)^2 (0.43)^3 = 0.260$	$100 \times (0.260) \approx 26$
3	34	${}^5C_3 (0.57)^3 (0.43)^2 = 0.342$	$100 \times (0.342) \approx 34$
4	22	${}^5C_4 (0.57)^4 (0.43) = 0.224$	$100 \times (0.224) \approx 22$
5	8	${}^5C_5 (0.57)^5 (0.43)^0 = 0.059$	$100 \times (0.059) \approx 6$

Example 1.69. In a Binomial distribution consisting of 5 independent trials, probability of and 2 success are 0.4096 and 0.2048 respectively. Find the parameter ' p ' of distribution.

Solution. Given $n = 5$

For Binomial distribution

$$P(X = x) = {}^nC_x p^x q^{n-x}, x = 0, 1, 2, 3, \dots, n$$

$$P(X = 1) = {}^nC_1 p^1 q^{n-1} = 0.4096$$

$$P(X = 2) = {}^nC_2 p^2 q^{n-2} = 0.2048$$

Dividing eqn. (ii) with eqn. (i)

$$\frac{{}^5C_2 p^2 q^3}{{}^5C_1 p^1 q^4} = \frac{10p}{5q} = \frac{0.4096}{0.2048}$$

$$\Rightarrow \frac{2p}{1-p} = 1/2$$

$$\Rightarrow 4p = 1-p$$

$$\Rightarrow p = 1/5$$

EXERCISE 1.4

1. During war, 1 ship out of 9 was sunk on an average in making a certain voyage. What is the probability that exactly 3 out of the convey of 6 ships would arrive safely?

[Ans. (i) $\frac{104}{9}$]

2. The incidence of occupational disease in an industry in such that the workers have a 20% chance of suffering from it. What is the probability that out of six workers chosen at random, four of them will suffer from the disease?

[Ans. (i) $\frac{1}{12}$]

3. The probability that a pen manufactured by a company will be defective is $\frac{1}{10}$. If 12 such pens are manufactured, find the probability that : (i) exactly 2 will be defective, (ii) none will be defective, (iii) at least two will be defective.

[Ans. (i) $\frac{3}{32}$, (ii) $\frac{7}{64}$, (iii) $\frac{1}{16}$]

4. A dice is thrown. If "getting an odd number" is a "success", what is the probability of : (i) 5 successes, (ii) at least 5 successes, (iii) almost 5 successes

[Ans. (i) $\frac{3}{32}$, (ii) $\frac{7}{64}$, (iii) $\frac{1}{16}$]

5. If on an average one ship in every ten is wrecked, find the probability that out of 5 ships expected to arrive, 4 at least will arrive safely.

[Ans. (i) $\frac{7}{16}$, (ii) $\frac{1}{16}$, (iii) $\frac{1}{16}$]

6. Five cards are drawn successively with replacement from a well-shuffled pack of 52 cards. Find the probability that : (i) all the five cards are spades, (ii) only 3 cards are spades, (iii) none is a spade

[Hints. Number of spades : 13] [Ans. (i) $\left(\frac{1}{4}\right)^5$, (ii) $90 \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^2$, (iii) $\left(\frac{3}{4}\right)^5$]

7. State reason to justify whether the following statement is true or false. "The mean of a Binomial distribution is 6 and standard deviation is 3".

[Ans. True]

8. A and B take turns in throwing dice, the first to throw 10 being the winner. If A throws first, then they have a change of winning as 12 : 11.

[Ans. True]

9. In a bombing action there is 50% chance that any bomb will strike target. Two direct hits are needed to destroy a target completely. How many bombs are required to be dropped to have 99% chance of better of completely destroying the target.

[Ans. 10]

10. Out of 800 families with 5 children each, how many would you expect to have : (i) 3 girls, (ii) 5 girls, (iii) either 2 or 3 boys? Assume equal probabilities for boys and girls.

[Ans. (i) 250, (ii) 25, (iii) 50]

BASIC PROBABILITY

12. The probability that a bulb produced by a factory will fuse after 150 days of use is $\frac{1}{20}$. Find the probability that out of 5 such bulbs : (i) None, (ii) not more than one, (iii) more than one, (iv) at least one, will fuse after 150 days of use.

[Ans. (i) $\left(\frac{19}{20}\right)^5$, (ii) $\frac{24}{20} \left(\frac{19}{20}\right)^4$, (iii) $1 - \frac{23}{20} \left(\frac{19}{20}\right)^4$, (iv) $1 - \left(\frac{19}{20}\right)^5$]

13. Four coins are tossed 160 times. The number of times r heads occur ($r = 0, 1, 2, 3, 4$) is given below:

r	0	1	2	3	4
No. of times	8	34	69	43	6

Fit a binomial distribution to this data on the hypothesis that coins are unbiased.

[Ans. $r = 0, 1, 2, 3, 4$
 $f(r) = 10, 40, 60, 40, 10$]

14. If successive trials are independent and the probability of success on any trial is p , shown that the first success occurs on the n th trial is

$$p(1-p)^{n-1}, n = 1, 2, 3, \dots$$

15. The mean of a Binomial distribution is 3 and variance is 4. Give your comments.

16. If the chance that one of the ten telephone lines is busy at an instant is 0.2.

(i) What is the chance that 5 of the lines are busy?

(ii) What is the probability that all the lines are busy? [Ans. (i) 0.02579 , (ii) 21.024×10^{-7}]

17. The following data are the number of seeds germinating out of 10 on damp filter for 80 sets of seeds. Fit a binomial distribution of this data :

x	0	1	2	3	4	5	6	7	8	9	10	Total
f	6	20	28	12	8	6	0	0	0	0	0	80

[Ans. $80(0.7825 + 0.2175)^{10}$]

18. Mark the correct answer.

- (a) The probability that a man hit a target is given as $\frac{1}{5}$. Then his probability of atleast one hit in 10 shots is

(i) $1 - \left(\frac{4}{5}\right)^{10}$ (ii) $\left(\frac{1}{5}\right)^{10}$ (iii) $1 - \left(\frac{1}{5}\right)^{10}$ (iv) None

- (b) 8 coins are tossed simultaneously. The probability of getting at least 6 heads is

(i) $\frac{57}{64}$ (ii) $\frac{229}{256}$ (iii) $\frac{7}{64}$ (iv) $\frac{37}{256}$

[Ans. (a) (i), (iv) (b) (ii)]

20. Fill in the blanks :

- The probability of getting number 5 exactly two times in five throws of an unbiased die is $\frac{1}{6^5}$.
- The probability of hitting a target is 5% and 5 shots are fired independently, the probability that the target will be hit at least once is $1 - (0.95)^5$.

[Ans. (a) $10 \left(\frac{5^5}{6^5} \right)$, (b) $1 - (0.95)^5$]

1.24. POISSON DISTRIBUTION

It is due to the French Mathematician S.D. Poisson (1781-1842) who published its derivation in 1837. It is a discrete probability distribution which has the following characteristics:

- It is the limiting form of the Binomial distribution as n becomes infinitely large i.e., $n \rightarrow \infty$ and p , the constant probability of success for each trial becomes indefinitely small i.e., $p \rightarrow 0$ in such a manner that $np = m$ remains a finite number.
- It consists of a single parameter m only. The entire distribution can be obtained once m is known. It has wide applications in physical, engineering and management sciences as well as economics, operations research and reliability technology.

Poisson distribution occurs when there are events which do not occur as outcomes of definite number of trials of an experiment but which occurs at random point of time and space wherein our interest is only in the number of occurrences of the events. Here we are concerned with processes taking place over continuous intervals of time such as arrival of a telephone calls at a switch board, or the passing of cars of an electric checking device. That is why it has important applications in queuing theory where we are interested, for example, in the number of aircraft arriving at an air field, the number of ships and trucks arriving to be unloaded at a receiving dock etc.

Some of the situations where the probability of the event remains very small and the Poisson distribution is applicable, are: (i) the occurrence of accidents in a factory in a given period, (ii) the number of defective articles produced by some factory in a fixed time period, (iii) the number of deaths in a city per year, (iv) the number of printing mistakes which page of the book, (v) the number of persons blind in a large city every year etc.

1.25. BINOMIAL APPROXIMATION TO POISSON DISTRIBUTION

We will derive Poisson distribution as a limiting case of Binomial distribution when $p \rightarrow 0$, $n \rightarrow \infty$ such that $np = m$ (a finite quantity). We know that in a Binomial distribution, the probability of successes is given by

$$P(r) = {}^nC_r q^{n-r} r^r = \frac{n(n-1)(n-2)\dots(n-r+1)}{r!} (1-p)^{n-r} p^r$$

on expanding nC_r and replacing q by $(1-p)$.

$$P(r) = \frac{n(n-1)(n-2)\dots(n-r+1)}{r!} \left(1 - \frac{m}{n}\right)^{n-r} \left(\frac{m}{n}\right)^r$$

since $np = m$ or $p = \frac{m}{n}$

or

$$P(r) = \frac{n(n-1)(n-2)\dots(n-r+1)}{r! m^r} \left(1 - \frac{m}{n}\right)^{n-r} \left(\frac{m}{n}\right)^r \cdot m^r$$

$$= \frac{n(n-1)(n-2)\dots(n-r+1)}{r!} \left(1 - \frac{m}{n}\right)^{n-r} \left(\frac{m}{n}\right)^r \times \left(\frac{n}{m}\right)^r$$

$$= \left(1 - \frac{m}{n}\right)^{n-r} \left(1 - \frac{m}{n}\right)^r \left(1 - \frac{m}{n}\right)^{-r} \left(1 - \frac{m}{n}\right)^r \times \left(\frac{n}{m}\right)^r$$

Now for a given value of r , when $n \rightarrow \infty$, we have

$$\lim_{n \rightarrow \infty} \left[\left(1 - \frac{m}{n}\right)^{n-r} \left(1 - \frac{m}{n}\right)^r \right] = 1,$$

$$\lim_{n \rightarrow \infty} \left(1 - \frac{m}{n}\right)^{n-r} = 1 \text{ and } \lim_{n \rightarrow \infty} \left(1 - \frac{m}{n}\right)^r = e^{-m}$$

As such in the limiting form

$$P(r) = \frac{e^{-m} \cdot m^r}{r!}$$

This is the probability of r successes for Poisson distribution. For $r = 0, 1, 2, 3, \dots$, we get the probabilities of 0, 1, 2, 3, ... successes as

$$P(0) = e^{-m}, P(1) = m e^{-m}, P(2) = \frac{m^2}{2!} e^{-m}$$

$$P(3) = \frac{m^3}{3!} e^{-m} \dots \text{and so on}$$

Note : 1. The sum of the probabilities $P(r)$ for $r = 0, 1, 2, 3, \dots$ is 1.

$$\Sigma P(r) = P(0) + P(1) + P(2) + P(3) + \dots$$

$$= e^{-m} \left(1 + m + \frac{m^2}{2!} + \frac{m^3}{3!} + \dots \right) = e^{-m} \times e^m = 1.$$

2. Poisson distribution possesses only one parameter ' m '. If we consider the length of interval as ' t ' instead of unit length, the average number of occurrences in ' t ' length of interval is ' mt '. Thus, the probability function in this situation is,

$$P(r) = \frac{e^{-mt} (mt)^r}{r!}$$

1.26. CONDITIONS UNDER WHICH POISSON DISTRIBUTION IS USED

- The random variable x should be discrete.
- A dichotomy exists, i.e., the happening of the event must be of two alternatives such as success and failure, occurrences and non-occurrence etc.
- It is applicable in those case where the number of trials n is very large and the probabilities of success p is very small but the mean $np = m$ is finite.

4. p should be very small (close to zero). If $p \rightarrow 0$, then the distribution is J-shaped and normality is not applicable. In other words, it is applicable to those cases where the probability of an event happening is very small.
5. Statistical independence is assumed. In other words, it is applicable to those cases where the happening of an event does not affect the happening of the other events.

1.27. MEAN OF POISSON DISTRIBUTION

Poisson distribution is $P(r) = \frac{e^{-m} m^r}{r!}$, $r = 0, 1, 2, \dots$

Mean, $(\mu) = \sum_{r=0}^{\infty} r \cdot P_r = P_1 X_1 + P_2 X_2 + P_3 X_3 + \dots + P_n X_n$

$$= 0e^{-m} + 1e^{-m} + 2 \frac{m^2 e^{-m}}{2!} + 3 \frac{m^3 e^{-m}}{3!} + 4 \frac{m^4 e^{-m}}{4!} + \dots$$

$$= me^{-m} + m^2 e^{-m} + \frac{m^3 e^{-m}}{2 \times 1} + \frac{m^4 e^{-m}}{3 \times 2 \times 1} + \dots$$

$$= me^{-m} \left(1 + m + \frac{m^2}{2 \cdot 1} + \dots \right)$$

$$\left(\because e^m = 1 + m + \frac{m^2}{2!} + \frac{m^3}{3!} + \dots \right)$$

$$= me^{-m} e^m$$

$$= me^{-m} + n = me^0 = m$$

$\therefore$ The mean of the Poisson distribution is m .

1.28. VARIANCE OF POISSON DISTRIBUTION

Variance $(\sigma^2) = (\sum X_i^2 P_i) - \mu^2$

Now

$$\sum_{r=0}^{\infty} r^2 P_r = 0e^{-m} + 1 \cdot me^{-m} + 2^2 \cdot \frac{m^2 e^{-m}}{2!} + 3^2 \cdot \frac{m^3 e^{-m}}{3!} + 4^2 \cdot \frac{m^4 e^{-m}}{4!} + \dots$$

$$= me^{-m} + 2m^2 e^{-m} + \frac{3m^3 e^{-m}}{2 \times 1} + \frac{4m^4 e^{-m}}{3 \times 2 \times 1} + \dots$$

$$= me^{-m} \left[1 + 2m + \frac{3m^2}{2 \times 1} + \frac{4m^3}{3 \times 2 \times 1} + \dots \right]$$

$$= me^{-m} \left[\left(1 + m + \frac{m^2}{2 \times 1} + \frac{m^3}{3 \times 2 \times 1} + \dots \right) + \left(m + \frac{2m^2}{2} + \frac{3m^3}{3 \times 2 \times 1} + \dots \right) \right]$$

$$= me^{-m} \left[e^m + m \left(1 + m + \frac{m^2}{2!} + \frac{m^3}{3!} + \dots \right) \right]$$

$$= me^{-m} (e^m + me^m) = m(m+1) e^{-m} e^m = m(m+1)$$

$$\text{Variance} = \sum r^2 P_r = \mu^2 m(m+1) - m^2 = m^2 + m - m^2 = m$$

$$\text{Standard deviation} = \sqrt{\text{Variance}} = \sqrt{m}$$

BASIC PROPERTIES

1.29. MODE OF POISSON DISTRIBUTION

Mode is a value of r at which $P(r)$ has maximum value, then

$$P(r) = P(r+1) \text{ and } P(r) = P(r-1)$$

$$\frac{e^{-m} m^r}{r!} \geq \frac{e^{-m} m^{r+1}}{(r+1)!} \Rightarrow \frac{r+1}{m} \geq 1$$

$$1e^{-m} \Rightarrow r+1 \geq m \Rightarrow r \geq m-1$$

$$m-1 \leq r$$

Similarly consider

$$P(r) \geq P(r-1) \text{ then we get } r \geq m$$

From eqns. (i) and (ii)

$$m-1 \leq r \leq m$$

Therefore the mode of the Poisson distribution lies between $(m-1)$ and m .

Constants of Poisson Distribution

$$\mu_1 = 0, \mu_2 = m, \mu_3 = m, \mu_4 = m + 3m^2$$

$$\beta_1 = \frac{\mu_1^2}{\mu_2} = \frac{m^2}{m} = \frac{1}{m}$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{m + 3m^2}{m^2} = 3 + \frac{1}{m}$$

1.30. RECURRENCE FORMULA FOR THE POISSON DISTRIBUTION

We have

$$P(r) = \frac{e^{-m} \cdot m^r}{r!}$$

and

$$P(r+1) = \frac{e^{-m} \cdot m^{r+1}}{(r+1)!}$$

Then

$$\frac{P(r+1)}{P(r)} = \frac{\frac{e^{-m} \cdot m^{r+1}}{(r+1)!}}{\frac{e^{-m} \cdot m^r}{r!}} = \frac{e^{-m} \cdot m^{r+1}}{e^{-m} \cdot m^r} \times \frac{r!}{(r+1)!} = \frac{m}{r+1}$$

$$\Rightarrow P(r+1) = \frac{m}{r+1} P(r); r = 0, 1, 2, \dots$$

which is the required recurrence formula for Poisson distribution. With this formula, we can find $P(1), P(2), P(3), P(4), \dots$ if $P(0)$ is given.

SOLVED EXAMPLES

Example 1.70. The number of telephone calls arriving on an internal switch board is 60 per hour. Find the probability that at the most 1 to 3 calls in a minute on the average. (Use $e^{-1.5} = 0.223$)

Solution. $m = \text{mean} = \frac{60}{60} = 1.5$. Obviously, X will follow Poisson distribution.

Now the probability that at the most 1 to 3 calls in one minute

$$= P(X=1) + P(X=2) + P(X=3)$$

where

$$P(X) = \frac{e^{-m} m^x}{x!}$$

$$P(X=1) = \frac{e^{-1.5} (1.5)^1}{1!} = 1.5 \times e^{-1.5} = 1.5 \times 0.223$$

$$P(X=2) = \frac{e^{-1.5} (1.5)^2}{2!} = 1.125 \times e^{-1.5} = 1.125 \times 0.223$$

$$P(X=3) = \frac{e^{-1.5} (1.5)^3}{3!} = 0.562 \times e^{-1.5} = 0.562 \times 0.223$$

$$\therefore \text{Required probability} = 0.233 (1.5 + 1.125 + 0.562) = 0.711$$

Example 1.71. Suppose a book of 585 pages contains 43 typographical errors. If these are randomly distributed throughout the book, what is the probability that 10 pages who are random, will be free from errors? (Use $e^{-0.735} = 0.4795$).

Solution. Here $p = \frac{43}{585} = 0.0735$ and $n = 10$.

$$m = np = 10 \times 0.0735 = 0.735$$

Clearly, p is very small and n is large.

So, it is a case of Poisson distribution.

Let X denote the number of errors in 10 pages.

Then,

$$P(X=r) = \frac{e^{-m} m^r}{r!} = \frac{e^{-0.735} \times (0.735)^r}{r!}$$

$$\therefore P(\text{no error}) = P(X=0) = \frac{e^{-0.735} \times (0.735)^0}{0!} = e^{-0.735} = 0.4795$$

Hence, the required probability is 0.4795.

Example 1.72. Average number of accidents on any day on a national highway is 1.8. Find the probability that the number of accidents are: (i) at least one, (ii) at most one. (Given $e^{-1.8} = 0.16529$).

Basic Probability

Solution. The probability function of the Poisson distribution is

$$P(X=r) = \frac{e^{-m} m^r}{r!} \quad (r=0, 1, 2, 3, \dots)$$

Given that $m = 1.8$

(i) $P(X \geq 1)$:

$$P(X \geq 1) = 1 - P(X \leq 1)$$

$$= 1 - P(X=0)$$

$$= 1 - \frac{e^{-m} m^0}{0!} = 1 - \frac{e^{-1.8} (1.8)^0}{0!}$$

$$= 1 - e^{-1.8} = 1 - 0.16529 = 0.8347$$

(ii) The probability the number of accidents at least one is 0.8347.

$$(ii) P(X \leq 1) = P(X=0) + P(X=1)$$

$$= \frac{e^{-m} m^0}{0!} + \frac{e^{-m} m^1}{1!} = \frac{e^{-1.8} (1.8)^0}{0!} + \frac{e^{-1.8} (1.8)^1}{1!}$$

$$= e^{-1.8} (1 + 1.8) = 0.4628$$

(iii) The probability that the number of accidents at most one is 0.4628.

Example 1.73. In a certain factory turning out razor blades, there is a small chance of 0.002 for any blade to be defective. The blades are supplied in packets of 10. Calculate the approximate number of packets containing no defective, one defective and two defective blades in a consignment of 10,000 packets. (Given $e^{-0.02} = 0.9802$).

Solution. Here $N = 10000$, $p = 0.02$, $n = 10$.

$$\text{Mean } (m) = np = 10 \times 0.002 = 0.02$$

Let r be the number of defective blades in a packet.

Let $P(r)$ be the number of packets containing r defective blades, then

$$P(r) = N \times \frac{e^{-m} m^r}{r!}$$

(i) $P(0)$ = Number of packets with no defective blades

$$= 10000 \times \left[0.9802 \times \frac{(0.02)^0}{0!} \right] = 10000 (0.9802) = 9802$$

(ii) $P(1)$ = Number of packets with 1 defective blade

$$= 1000 \left[0.9802 \times \frac{(0.02)^1}{1!} \right] = 10000 [0.9802 \times 0.02]$$

$$= 10000 (0.019604) = 196$$

(iii) Number of packets with one defective blade = 196.04 or 196

$$(iii) P(2) = \text{Number of packets with 2 defective blades} \\ = 10000 \left[\frac{0.9802 \times \frac{(0.02)^2}{2!}}{2!} \right] = 10000 \left[\frac{0.9802 \times 0.0004}{2} \right]$$

$$= 10000 (0.00019604) = 1.96 \\ = 1.96 \text{ or } 2.$$

$\therefore$ Number of packets with two defective blades = 1.96 or 2.

Example 1.74. If the variance of the Poisson distribution is 2, find the probabilities for 1, 3, 4 from the recurrence relation of the Poisson distribution. Also find $P(X \geq 4)$. (Use $e^{-2} = 0.1353$)

Solution. Here variance = $m = 2$, $P(0) = e^{-2} = 0.1353$ (given)

$$\text{We know that } P(r+1) = \frac{m}{r+1} P(r) = \frac{2}{r+1} P(r)$$

Putting $r = 0, 1, 2, 3$ in (i) we get

$$P(1) = \frac{2}{0+1} \times P(0) = 2e^{-2} = 2 \times 0.1353 = 0.2706$$

$$P(2) = \frac{2}{2} \times P(1) = 0.2706$$

$$P(3) = \frac{2}{3} \times P(2) = \frac{2}{3} \times 0.2706 = 0.1804$$

$$P(4) = \frac{2}{4} \times P(3) \times \frac{1}{2} \times P(3) = 0.0902$$

$$\text{Now, } P(X \geq 4) = 1 - [P(0) + P(1) + P(2) + P(3)] \\ = 1 - [0.1353 + 0.2706 + 0.2706 + 0.1804] = 0.1431$$

Example 1.75. If a random variable X follows a Poisson distribution such that $P(X=2) = 9$, $P(X=4) = 90$, $P(X=6)$, find the mean and variance of X .

Solution. By Poisson distribution, we have

$$P(X=r) = \frac{e^{-m} m^r}{r!}$$

$$\text{Now } P(X=2) = 9, P(X=4) = 90, P(X=6)$$

$$\Rightarrow \frac{e^{-m} m^4}{4!} = \frac{9 \cdot e^{-m} m^4}{4!} + \frac{90 \cdot e^{-m} m^6}{6!}$$

$$\Rightarrow \frac{3}{4} m^3 + \frac{1}{4} m^4 = 1 \Rightarrow m^4 + 3m^2 - 4 = 0 \\ \Rightarrow (m^2 + 4)(m^2 - 1) \Rightarrow m^2 = 1 \Rightarrow m = 1$$

Hence, mean = 1 and variance = 1

[$\because m > 0$ and $m^2 + 4 > 0$]
[$\therefore$ Mean = Variance]

Example 1.76. An insurance company has discovered that only 0.1% of the population is involved in a certain type of accident every year. If its 1000 policy-holders are selected at random from the population, what is the probability that not more than 5 of its clients are involved in such accident next year? (Use $e^{-1} = 0.3668$)

Solution. Here, n = total number of policy-holders = 1000

Let p = probability of a person being involved in an accident = $\frac{0.1}{100} = 0.001$

$$m = np = 1000 \times 0.001 = 1$$

Here, n is large and p is very small.

So, Poisson distribution can be used to find the required probability.

Let X be the Poisson variable with $m = 1$

$$\therefore P(X=r) = \frac{e^{-m} m^r}{r!} = \frac{1^r e^{-1}}{r!} = \frac{e^{-1}}{r!} \quad [\because m=1]$$

Required probability = $P(X \leq 5)$

$$= P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5)$$

$$= \left(\frac{e^{-1}}{0!} + \frac{e^{-1}}{1!} + \frac{e^{-1}}{2!} + \frac{e^{-1}}{3!} + \frac{e^{-1}}{4!} + \frac{e^{-1}}{5!} \right)$$

$$= e^{-1} \left(1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \frac{1}{120} \right)$$

$$= 0.368 \times \frac{326}{120} = \frac{3.68 \times 3.26}{120} = 0.9997$$

Hence, the required probability is 0.9997.

Example 1.77. Fit a Poisson distribution on the following :

x	0	1	2	3	4
f	192	100	24	3	1

(Given that $e^{-0.5} = 0.6065$)

Solution.

$$P(r) = \frac{e^{-m} m^r}{r!}$$

m = mean of the distribution

$$= \frac{0 \times 192 + 1 \times 100 + 2 \times 24 + 3 \times 3 + 4 \times 1}{192 + 100 + 24 + 3 + 1} = \frac{161}{320} = 0.5 \text{ (approx.)}$$

$$P(0) = \frac{e^{-0.5} (0.5)^0}{0!} = 0.6065 \text{ i.e., } f = 320 \times 0.6065 = 194 \text{ (approx.)}$$

$$P(1) = \frac{e^{-0.5}(0.5)^0}{1!} = 0.30325 \text{ i.e., } f = 320 \times 0.30325 \times 0.30325 = 97 \text{ (approx)}$$

$$P(2) = \frac{e^{-0.5}(0.5)^1}{2!} = 0.07581 \text{ i.e., } f = 320 \times 0.07581 = 24 \text{ (approx)}$$

$$P(3) = \frac{e^{-0.5}(0.5)^2}{3!} = 0.0126 \text{ i.e., } f = 320 \times 0.0126 = 4 \text{ (approx)}$$

$$P(4) = \frac{e^{-0.5}(0.5)^3}{4!} = 0.0016 \text{ i.e., } f = 320 \times 0.0016 = 0.512 \text{ or } 1 \text{ (approx)}$$

As total number of trials = 320
We have the approximate value as obtained by Poisson distribution as :

x	0	1	2	3	4
f	194	97	24	4	1

Example 1.78. A car hire firm has two cars which it hires out day by day. The number of demands for a car on each day is distributed as poisson distribution with mean 1.5. Calculate proportion, of days on which (i) neither car is used (ii) some demand is refused.

Solution. For Poisson distribution

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}; \quad x = 0, 1, 2, 3, \dots$$

Now

$$\lambda = 1.5$$

(i) The probability that there is no demand

$$P(X=0) = \frac{e^{-1.5}(1.5)^0}{0!} = e^{-1.5} = 0.223$$

The proportion of days on which neither car is used = 0.2231 = 22.31%

(ii) The probability that the demand is refused only when demand is more than 2 cars.

$$P(X > 2) = 1 - P(X \leq 2)$$

$$= 1 - [P(X=0) + P(X=1) + P(X=2)]$$

$$= 1 - \left\{ \frac{e^{-1.5}(1.5)^0}{0!} + \frac{e^{-1.5}(1.5)^1}{1!} + \frac{e^{-1.5}(1.5)^2}{2!} \right\}$$

$$= 0.1912$$

The proportion of days on which some demand is refused = 0.1912 = 19.12%

EXERCISE 1.5

- If a random variable has a Poisson distribution such that $P(1) = P(2)$, find:
 - Mean of the distribution
 - $P(4)$

[Ans. (i) 2, (ii) 3]

BASIC PROBABILITY

- A certain screw making machine produces on average 2 defective screws out of 100, and packs them in boxes of 500. Find the probability of contain 15 defectives screws. [Ans. 0.015]
- Six coins are tossed 1600 times. Use Poisson's distribution to find an expression of getting heads x times. [Ans. $P(X=x) = \frac{e^{-25}(25)^x}{x!}$]

- In sampling a large number of parts manufactured by machine, the mean number of defectives in a sample of 20 is 2. Out of 1000 such sample, how many would be expected to contain at least 3 defective parts? [Ans. 324]

- The probability that a mean aged 50 years will die within a year is 0.01125. What is the probability that of 12 such men at least 11 will reach their fifty first birthday. [Ans. 0.9916]

- Data was collected over a period of 10 years, showing number of deaths from horse kicks in each of the 200 army corps. The distribution of deaths was as following:

Number of deaths	0	1	2	3	4	Total
Frequency	109	65	22	3	1	200

Fit a Poisson distribution to the data the calculate the theoretical frequencies.

$$[\text{Ans. } 108.7, 109.20, 4.1]$$

- Find the expectation and variance of a variable X , if the probability $P(x, k) = \frac{e^{-m} m^k}{k!}$ for $k = 0, 1, 2, \dots, \infty$ (m is positive real number). [Ans. m, m]

- A manufacturer knows that the condensers he makes contain on an average 1% of defective. He packs them in boxes of 100. What is the probability that a box packed at random will contains 3 or more faulty condensers. [Ans. 0.08025]

- Fit a Poisson distribution of the following data and calculate theoretical frequencies:

Deaths	0	1	2	3	4
Frequency	122	60	15	2	1

$$[\text{Ans. Deaths: } 0 \quad 1 \quad 2 \quad 3 \quad 4 \\ \text{Frequency: } 121 \quad 61 \quad 15 \quad 2 \quad 1]$$

- If the probability that an individual suffers a bad reaction from a certain injection is 0.001. Find the probability that out of 2000 individual: (i) exactly 3, (ii) more than 2 individuals will suffer a bad reaction. [Ans. (i) 0.18, (ii) 0.32]

- An insurance company finds that 0.005% of the population dies from a certain kind of accident each year. What is the probability that the company must pay off no more than 3 of 10,000 insured risks against such accident in a given year? [Ans. 0.0016]

- Show that in a Poisson distribution with unit mean, and the mean deviation about the mean is $\frac{2}{e}$ times the standard deviation. [Hint: Mean deviation about the mean = $\sum p(r) |r - m|$ and $r = 0, 1, 2, \dots$]

1.36. MULTINOMIAL DISTRIBUTION

The distribution can be regarded as a generalisation of Binomial distribution.

When there are more than two mutually exclusive outcomes of a trial, the observations lead to multinomial distribution. Suppose $E_1, E_2, \dots, E_k$ are k mutually exclusive and exhaustive outcomes of a trial with the respective probabilities $p_1, p_2, \dots, p_k$.

The probability that E_1 occurs x_1 times, E_2 occurs x_2 times... and E_k occurs x_k times in n independent observations, is given by

$$P(x_1, x_2, \dots, x_k) = \frac{n!}{x_1! x_2! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}$$

where $\sum x_i = n$ and c is the number of permutation of the events $E_1, E_2, \dots, E_k$.

To determine c , we have to find the number of permutation of n objects of which x_1 are of kind x_1 , x_2 of another kind, ..., and x_k of the k th kind which is given by

$$c = \frac{n!}{x_1! x_2! \dots x_k!}$$

Hence,

$$p(x_1, x_2, \dots, x_k) = \frac{n!}{x_1! x_2! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}, \quad 0 \leq x_i \leq n$$

$$= \frac{n!}{\prod_{i=1}^k x_i!} \cdot \prod_{i=1}^k p_i^{x_i}, \quad \sum_{i=1}^k x_i = n$$

which is the required probability function of the multinomial distribution. It is so called since the general term in the multinomial expansion

$$(p_1 + p_2 + \dots + p_k)^n, \quad \sum_{i=1}^k p_i = 1$$

Since total probability is 1, we have

$$\sum_x p(x) = \sum_x \left[\frac{n!}{x_1! x_2! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k} \right]$$

$$= (p_1 + p_2 + \dots + p_k)^n = 1.$$

1.37. MOMENTS OF MULTINOMIAL DISTRIBUTION

The moment generating function is given by

$$M_X(t) = M_{X_1, X_2, \dots, X_k}(t_1, t_2, \dots, t_k) = E \left[\exp \left\{ \sum_{i=1}^k t_i X_i \right\} \right]$$

$$= \sum_x \left[\frac{n!}{x_1! x_2! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k} \exp \left\{ \sum_{i=1}^k t_i x_i \right\} \right]$$

$$= \sum_x \left[\frac{n!}{x_1! x_2! \dots x_k!} (p_1 e^{t_1})^{x_1} \dots (p_k e^{t_k})^{x_k} \right]$$

$$= (p_1 e^{t_1} + p_2 e^{t_2} + \dots + p_k e^{t_k})^n$$

$$x = (x_1, x_2, \dots, x_k)$$

$$M_{X_i}(t_i) = M_X(t_1, 0, \dots, 0) = (p_1 e^{t_1} + p_2 + p_3 + \dots + p_k)^n$$

$$= [(1 - p_1) + p_1 e^{t_1}]^n \quad \left(\because \sum_{i=1}^k p_i = 1 \right)$$

$$X_i \sim B(n, p_1)$$

[By uniqueness theorem of pdf]

Similarly, we shall get,

$$X_i \sim B(n, p_i), \quad i = 1, 2, \dots, k$$

$$\Rightarrow E(X_i) = np_i \text{ and } \text{Var } X_i = np_i(1 - p_i), \quad i = 1, 2, \dots, k$$

$$E(X_i X_j) = \left[\frac{\partial^2 M}{\partial t_i \partial t_j} \right]_{t=0}, \quad i \neq j$$

$$= [np_i e^{t_i} (n-1) p_j e^{t_j} + \dots + p_i e^{t_i} p_j e^{t_j}]_{t=0}$$

$$= n(n-1)p_i p_j$$

$$\text{Cov}(X_i, X_j) = E(X_i X_j) E(X_i) E(X_j) = n(n-1)p_i p_j - n^2 p_i p_j = -np_i p_j$$

$$\therefore \rho(X_i, X_j) = \frac{\text{Cov}(X_i, X_j)}{\sigma_{X_i} \sigma_{X_j}} = \frac{-np_i p_j}{\sqrt{np_i(1-p_i)np_j(1-p_j)}}$$

$$= - \left[\frac{p_i p_j}{(1-p_i)(1-p_j)} \right]^{1/2}$$

SOLVED EXAMPLES

Example 1.87. n balls are drawn with replacement from an urn containing ' a ' white and ' b ' black balls. Show that variance of the number of white balls drawn is $\frac{nab}{(a+b)^2}$.

Solution. p = probability of drawing a white ball = $\frac{a}{a+b}$

$$\therefore q = 1 - p = 1 - \frac{a}{a+b}$$

$$\text{Hence, variance} = np(1-p) = n \cdot \frac{a}{a+b} \left(1 - \frac{a}{a+b} \right) = \frac{nab}{(a+b)^2}.$$

Example 1.88. An urn contain 5 red, 2 white, 3 black balls. If 5 balls are drawn, what is the probability of obtaining 3 red, 1 white and 1 black ball: (i) without replacement, (ii) with replacement.

Solution. (i) When they are not replaced $p = \frac{5C_3 \times 2C_1 \times 3C_1}{10C_5} = \frac{5}{21}$

(ii) When they are replaced

$$p_1(\text{drawing a red ball}) = \frac{5}{10}$$

$$p_2(\text{drawing a white ball}) = \frac{2}{10}$$

$$p_3 \text{ (drawing a black ball)} = \frac{3}{10}$$

$$\therefore \text{Prob. (3 red, 1 white, 1 black)} = \frac{5!}{3!1!1!} \left(\frac{1}{2}\right)^3 \left(\frac{1}{5}\right)^1 \left(\frac{3}{10}\right)^1$$

$$= \frac{3}{20}$$

Example 1.89. A bag contains 5 red balls, 4 white balls and 3 blue balls. A ball is drawn random from the box, its colour is noted, and then the balls is replaced. Find the probability out of 6 balls selected in this manner 3 are red, 2 are white and 1 is blue.

Solution. P (red at any drawing) = $\frac{5}{12}$

$$P \text{ (white at any drawing)} = \frac{4}{12}$$

$$\text{and } P \text{ (blue at any drawing)} = \frac{3}{12}$$

$$\text{Then, } P(3 \text{ red, 2 white, 1 blue}) = \frac{6!}{3!2!1!} \left(\frac{5}{12}\right)^3 \left(\frac{4}{12}\right)^2 \left(\frac{3}{12}\right)^1$$

$$= \frac{625}{5184}$$

Example 1.90. The trinomial distribution of two r.v.'s X and Y is given by:

$$f_{X,Y}(x,y) = \frac{n!}{x!y!(n-x-y)!} p^x q^y (1-p-q)^{n-x-y}$$

for $x, y = 0, 1, 2, \dots, n$ and $x+y \leq n$, where $0 \leq p, 0 \leq q$ and $p+q \leq 1$

(i) Find the marginal distributions of X and Y

(ii) Find the conditional distributions of X and Y and obtain $E(Y/X=x)$ and $E(X/Y=y)$

(iii) Find the correlation coefficient between X and Y .
Solution. The joint m.g.f. of X and Y is given by:

$$M_{X,Y}(t_1, t_2) = E(e^{t_1 X + t_2 Y}) = \sum_{x=0}^n \sum_{y=0}^{n-x} (pe^{t_1})^x (qe^{t_2})^y (1-p-q)^{n-x-y}$$

$$= [pe^{t_1} + qe^{t_2} + (1-p-q)]^n$$

$$M_X(t_1) = M(t_1, 0) = (1-p + pe^{t_1})^n \Rightarrow X \sim B(n, p)$$

$$M_Y(t_2) = M(0, t_2) = \{(1-q) + qe^{t_2}\}^n \Rightarrow Y \sim B(n, q)$$

Observe that $M(t_1, t_2) \neq M(t_1, 0) \times M(0, t_2) \Rightarrow X$ and Y are not independent.

(ii) The conditional distribution of X given $Y=y$ is given by:

$$f_{X|Y}(x,y) = \frac{f_{XY}(x,y)}{f_Y(y)} = \frac{f_{XY}(x,y)}{{}^nC_y q^y (1-q)^{n-y}} \quad [\because Y \sim B(n, q)]$$

$$= \frac{(n-y)!}{x!(n-y-x)!} \left(\frac{p}{1-q}\right)^x \left(\frac{1-p-q}{1-q}\right)^{n-y-x}$$

$$= \binom{n-y}{x} \left(\frac{p}{1-q}\right)^x \left(1 - \frac{p}{1-q}\right)^{n-y-x}; \quad x=0, 1, \dots, n$$

$$\Rightarrow X|Y=y \sim B(n-y, p/(1-q))$$

$$\Rightarrow E(X|Y=y) = (n-1) \cdot p/(1-q)$$

Similarly, we shall get

$$f_{Y|X}(x,y) = \frac{f_{XY}(x,y)}{f_X(x)} = \frac{f_{XY}(x,y)}{{}^nC_x p^x (1-p)^{n-x}} \quad [\because X \sim B(n, p)]$$

$$= \binom{n-x}{y} \left(\frac{q}{1-p}\right)^y \left(1 - \frac{q}{1-p}\right)^{n-x-y}; \quad y=0, 1, \dots, n$$

$$\Rightarrow Y|X=x \sim B(n-x, q/(1-p))$$

$$\Rightarrow E[Y|X=x] = (n-x)q/(1-p)$$

(iii) Correlation Coefficient ρ_{XY} :

$$\text{Since } X \sim B(n, p), E(X) = np, \text{Var } X = np(1-p)$$

$$Y \sim B(n, q), E(Y) = nq, \text{Var } Y = nq(1-q)$$

$$E(XY) = \frac{\partial^2 M(t_1, t_2)}{\partial t_1 \partial t_2} \bigg|_{t_1=t_2=0} = n(n-1)pq$$

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = n(n-1)pq - n^2pq = -npq$$

$$\therefore \rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{-npq}{\sqrt{np(1-p)} \sqrt{nq(1-q)}} = - \left[\frac{pq}{(1-p)(1-q)} \right]$$

Note. Here $p+q \neq 1$.

Example 1.91. If $X_1, X_2, \dots, X_k$ are k independent Poisson variates with parameters $\lambda_1, \lambda_2, \dots, \lambda_k$ respectively, prove that the conditional distribution $P(X_1 \cap X_2 \cap \dots \cap X_k | X)$, where $X = X_1 + X_2 + \dots + X_k$ is fixed, is multinomial.

Solution. $P[X_1 \cap X_2 \cap \dots \cap X_k | X=n]$

$$= P[X_1 = r_1 \cap X_2 = r_2 \cap \dots \cap X_k = r_k | X=n]$$

$$= \frac{P[X_1 = r_1 \cap X_2 = r_2 \cap \dots \cap X_k = r_k \cap X=n]}{P(X=n)}$$

EXERCISE 1.7

1. Define multinomial distribution and write its probability function. Also, obtain its m.g.f. and deduce mean, variance and covariance.

2. A die is tossed 12 times, find the probability of getting 1, 2, 3, 4, 5 and 6 points exactly 2 times each.

$$\left(\text{Ans. } \frac{12!}{(2!)^6} \left(\frac{1}{6} \right)^{2 \times 6} = 0.003444 \right)$$

3. An urn contains 12 black, 9 red, 6 white and 3 blue balls. 10 balls are drawn at random: (i) without replacement, (ii) with replacement. What is the probability that there are 2 black, 3 red, 5 white and no blue balls in each case? [Ans. (i) 0.001, (ii) 0.087]

4. If $X_1, X_2, \dots, X_k$ have a multinomial distribution with the parameters n and p_i ($i = 1, 2, \dots, k$) with $\sum p_i = 1$, obtain the joint probability

$$P(X_1 = x_1 \cap X_2 = x_2 \cap \dots \cap X_k = x_k)$$

Obtain the corresponding moment generating function. Hence, or otherwise show that $E(X_i) = np_i$, $V(X_i) = np_i(1 - p_i)$

and $\text{Cov}(X_i, X_j) = -np_i p_j$ ($i \neq j$).

5. If the random variables $X_1, X_2, \dots, X_k$ have a multinomial distribution, show that the marginal distribution of X_i is a binomial distribution with the parameters n and p_i , with $i = 1, 2, \dots, k$.

6. For the trinomial distribution of two r.v.'s X and Y given by:

$$f(x, y) = \frac{n!}{x! y! (n - x - y)!} p_1^x p_2^y p_3^{n-x-y}$$

where x and y are non-negative integers with $x + y \leq n$ and p_1, p_2, p_3 are proper positive fractions with $p_1 + p_2 + p_3 = 1$, and

$$f(x, y) = 0, \text{ otherwise}$$

Show that (i) $X \sim B(n, p_1)$ and $Y \sim B(n, p_2)$

(ii) $X | (Y = y) \sim B(n - y, p_1/(1 - p_2))$ and $Y | (X = x) \sim B(n - x, p_2/(1 - p_1))$

(iii) $\rho(x, y) = -[p_1/p_2(1 - p_1)(1 - p_2)]^{1/2}$.

7. Four tickets marked 00, 01, 10, 11 respectively are placed in a bag. A set is drawn at random five times, being replaced each time. Find the probability that the sum of the numbers on the tickets thus drawn is 23. [25/256]

8. Write down the probability functions of Binomials and Trinomial distribution if two and three variables are considered respectively.

$$[\text{Ans. } P(X_1 = x_1, X_2 = x_2) = \frac{n!}{x_1! x_2!} p_1^{x_1} p_2^{x_2}, p_1 + p_2 = 1, x_1, x_2 = 0, 1, 2, \dots, n; x_1 + x_2 = n]$$

$$P(X_1 = x_1, X_2 = x_2, X_3 = x_3) = \frac{n!}{x_1! x_2! x_3!} p_1^{x_1} p_2^{x_2} p_3^{x_3}, p_1 + p_2 + p_3 = 1;$$

$$x_1, x_2, x_3 = 0, 1, 2, \dots, n; x_1 + x_2 + x_3 = n]$$

9. Suppose that unbiased die has its faces renumbered so that a_i appears on one face, a_j appears on two faces and a_3 appears on the remaining three faces. Obtain the probability distribution of the random variables X_1, X_2, X_3 when the die is thrown 5 times and X_i = the number of times a_i appears, for $i = 1, 2, 3$. Also describe the sample space.

Ans. $P(X_1, X_2, X_3) = \frac{5!}{x_1! x_2! x_3!} \left(\frac{1}{6}\right)^{x_1} \left(\frac{2}{6}\right)^{x_2} \left(\frac{3}{6}\right)^{x_3}$, for $i = 1, 2, 3; x_i = 1, 2, 3, 4, 5; \sum_{i=1}^3 x_i = 5$

Sample space $S = \{(5, 0, 0), (4, 1, 0), (3, 2, 0), (2, 3, 0), (1, 4, 0), (0, 5, 0), (4, 0, 1), (3, 1, 1), (2, 2, 1), (1, 3, 1), (0, 4, 1), (3, 0, 2), (2, 1, 2), (1, 2, 2), (0, 3, 2), (2, 0, 3), (1, 1, 3), (0, 2, 3), (1, 0, 4), (0, 0, 5)\}$

1.38. MARKOV'S INEQUALITY

If X is a non-negative random variable, then for any $a > 0$

$$P[X \geq a] \leq \frac{E[X]}{a}$$

Proof. Let X be a continuous r.v. with p.d.f. $f(x)$. Then

$$\begin{aligned} E[X] &= \int_0^{\infty} xf(x) dx = \int_0^a xf(x) dx + \int_a^{\infty} xf(x) dx \\ &\geq \int_a^{\infty} xf(x) dx \geq \int_a^{\infty} af(x) dx = a \int_a^{\infty} f(x) dx = a P[X \geq a] \end{aligned}$$

$$\therefore \frac{E[X]}{a} \geq P[X \geq a] \leq \frac{E[X]}{a}$$

A generalized version of Markov's inequality is the following:

1.39. GENERALIZED MARKOV'S INEQUALITY

If X be a random variable and $g(\cdot)$ a non-negative function with R as its domain, then

$$P[g(x) \geq k] \leq \frac{E[g(X)]}{k}, \text{ for every } k > 0.$$

Proof. Suppose X is a continuous random variable with probability density function $f_X(\cdot)$. Then

$$\begin{aligned} E[g(X)] &= \int_{-\infty}^{\infty} g(x) f_X(x) dx \\ &= \int_A g(x) f_X(x) dx + \int_B g(x) f_X(x) dx, \\ \text{where } A &= \{x : g(x) \geq k\}, B = \{x : g(x) < k\}. \end{aligned}$$

$$\therefore E[g(X)] \geq \int_A g(x) f_X(x) dx, \text{ since } g(x) f_X(x) \geq 0 \text{ for } x \in R$$

$$\begin{aligned} &\geq \int_A kf_X(x) dx, \text{ since } g(x) \geq k \text{ for } x \in A \\ &= k \int_A f_X(x) dx = k P[g(X) \geq k]. \end{aligned}$$

$$\therefore \frac{1}{k} E[g(X)] \geq P[g(X) \geq k], \text{ since } k > 0.$$

$$\text{Hence, } P[g(X) \geq k] \leq \frac{E[g(X)]}{k}, \text{ for } k > 0.$$

$$\text{Corollary. } P[g(X) < k] \geq 1 - \frac{E[g(X)]}{k}$$

Proof. We have $P[g(X) < k] + P[g(X) \geq k] = 1$

$$\therefore P[g(X) < k] = 1 - P[g(X) \geq k] \geq 1 - \frac{E[g(X)]}{k}.$$

1.40. CHEBYSHEV'S INEQUALITY

Let X be a random variable with $E[X] = \mu$ and $\text{var}[X] = \sigma^2$.

Then for any constant $c > 0$,

$$P\{|X - \mu| \geq c\} \leq \sigma^2/c^2 \text{ or } P\{|X - \mu| < c\} \geq 1 - (\sigma^2/c^2).$$

Proof. By Markov's inequality, we have

$$P\{(X - \mu)^2 \geq c^2\} \leq \frac{E[(X - \mu)^2]}{c^2} = \frac{\sigma^2}{c^2}.$$

$$\text{Hence, } P\{|X - \mu| \geq c\} \leq \sigma^2/c^2, \because x^2 \geq a^2 \Leftrightarrow |x| \geq a \text{ for } a > 0]$$

To prove the second part, we notice that

$$P\{|X - \mu| < c\} + P\{|X - \mu| \geq c\} = 1$$

$$\Rightarrow P\{|X - \mu| < c\} = 1 - P\{|X - \mu| \geq c\} \geq 1 - (\sigma^2/c^2).$$

σ^2/c^2 is called Chebyshev's inequality bound for $P\{|X - m| \geq c\}$.

Remark 1. Chebyshev inequality can be restated as:

$$P\{|X - E[X]| < c\} \geq 1 - \frac{1}{c^2} \text{ var}[X].$$

Remark 2. If X is a random variable with finite variance, then

$$P\{|X - \mu| \geq r\sigma\} \geq \frac{1}{r^2} \text{ or } P\{|X - \mu| < r\sigma\} \geq 1 - \frac{1}{r^2}, \text{ for each } r > 0.$$

The above results follow by taking $c = r\sigma$ in (i).

Remark 3. From the above inequality, we have

$$P[\mu - \sigma < X < \mu + \sigma] \geq 1 - \frac{1}{r^2}.$$

It follows that the probability that X falls within r σ units of μ is greater than or equal to $1 - \frac{1}{r^2}$.

Remark 4. The Chebyshev's inequality gives a bound for the probability of an event given in terms of a random variable and its mean and variance. Further this bound does not depend on the distribution of X .

SOLVED EXAMPLES

Example 1.92. If X is a random variable such that $E[X] = 3$ and $E[X^2] = 13$, use the Chebyshev inequality to determine the lower bound for $P[-2 < X < 8]$.

Or

If X is a variate such that $E[X] = 3$, $E[X^2] = 13$, show that

$$P[-2 < X < 8] \geq 21/25.$$

Solution. We know $\text{var}[X] = E[X^2] - \{E[X]\}^2$

$$= 13 - 9 = 4 \Rightarrow \sigma = 2.$$

Now

$$P[-2 < X < 8] = P[-2 - 3 < X - 3 < 8 - 3]$$

$$= P[-5 < X - 3 < 5]$$

$$= P[|X - 3| < 5] \geq 1 - (2^2/\sigma^2).$$

Hence, $P[-2 < X < 8] \geq 21/25$ i.e., $21/25$ is a lower bound for $P[-2 < X < 8]$ by Chebyshev's inequality.

Example 1.93. Variate X takes the values $-1, 1, 3, 5$ with associated probability $\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}$.

Compute $p = P[|X - 3| \geq 1]$ directly and find an upper bound to p by using Chebyshev's inequality.

Solution.

$$E[X] = -1 \cdot \frac{1}{6} + 1 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} = 3,$$

$$E[X^2] = 1 \cdot \frac{1}{6} + 1 \cdot \frac{1}{6} + 9 \cdot \frac{1}{6} + 25 \cdot \frac{1}{6} = \frac{43}{3}.$$

$$\therefore \sigma^2 = \text{var}[X] = E[X^2] - \{E[X]\}^2 = \frac{43}{3} - 9 = \frac{16}{3}.$$

By Chebyshev's inequality, we have

$$P = P[|X - 3| \leq 1] \geq \frac{\sigma^2}{c^2} = \frac{16}{3}. \quad (\because c=1)$$

Hence, an upper bound to p is $16/3$.

$$\begin{aligned} P &= 1 - P[|X - 3| < 1] = 1 - P[-1 < X - 3 < 1] \\ &= 1 - P[-2 < X < 4] = 1 - P[X = 3] = 1 - \frac{1}{6} = \frac{5}{6}. \end{aligned}$$

BASIC PROBABILITY

Example 1.94. If X is the number scored in a throw of a fair die, show that the Chebyshev's inequality gives $P[|X - \mu| > 2.5] < 0.47$, while the actual probability is zero.

Solution. We have $\mu = E[X] = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \frac{7}{2} = 3.5$,

$$E[X^2] = \frac{1}{6}(1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2) = \frac{91}{6}.$$

$$\therefore \sigma^2 = \text{var}[X] = E[X^2] - \{E[X]\}^2 = \frac{91}{6} - \frac{49}{4} = \frac{35}{12}.$$

By Chebyshev's inequality, we have

$$P[|X - \mu| > 2.5] \leq \frac{35}{12(2.5)^2} = 0.47.$$

Actual probability

$$\begin{aligned} &= P[|X - \mu| > 2.5] = 1 - P[|X - 3.5| \leq 2.5] \\ &= 1 - P[-2.5 \leq X \leq 2.5 + 3.5] \\ &= 1 - P[1 \leq X \leq 6] = 1 - 1 = 0. \end{aligned}$$

Example 1.95. A discrete variate X can assume only the values $x = 1, 2, 3, \dots$ with probabilities

2^{-x} . Show that Chebyshev's inequality gives $P[|X - 2| \leq 2] > \frac{1}{2}$, while the actual probability is $15/16$.

Solution.

$$E[X] = \sum_{x=1}^{\infty} \frac{x}{2^x} = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \frac{4}{2^4} + \dots$$

$$= \frac{1}{2}(1 + 2A + 3A^2 + 4A^3 + \dots), \quad (A = \frac{1}{2})$$

$$= \frac{1}{2}(1 - A)^{-2} = 2.$$

$$E[X^2] = \sum_{x=1}^{\infty} \frac{x^2}{2^x}$$

$$= \frac{1}{2}[1 + 4A + 9A^2 + \dots] = \frac{1}{2}(1 + A)(1 - A)^{-3} = 6.$$

$$\therefore \sigma^2 = \text{var}[X] = E[X^2] - \{E[X]\}^2 = 6 - 4 = 2.$$

By Chebyshev's inequality, we have

$$P[|X - 2| \leq 2] > 1 - \sigma^2/c^2 = 1 - \frac{1}{2} \quad (\because c=2, \sigma^2=2)$$

$$\therefore P[|X - 2| \leq 2] > \frac{1}{2}.$$

The actual probability is given by

$$P[|X - 2| \leq 2] = P[-2 \leq X - 2 \leq 2] = P[0 \leq X \leq 4]$$

$$\begin{aligned}
 &= P[1] + P[2] + P[3] + P[4], P(x) = \frac{1}{2^x} \\
 &= \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^4 = \frac{15}{16}.
 \end{aligned}$$

Example 1.96. Let X have the *pdf*:

$$\begin{aligned}
 f(x) &= \frac{1}{2\sqrt{3}}, \quad -\sqrt{3} < x < 3 \\
 &= 0, \quad \text{elsewhere.}
 \end{aligned}$$

Find the actual probability $P[|X - \mu| \leq \frac{3}{2}\sigma]$ and compare it with the upper bound obtained by Chebychev's inequality.

Solution. We have

$$E[X] = \int_{-\sqrt{3}}^{\sqrt{3}} \frac{x}{2\sqrt{3}} dx = 0,$$

$$E[X^2] = \int_{-\sqrt{3}}^{\sqrt{3}} \frac{x^2}{2\sqrt{3}} dx = \frac{1}{6\sqrt{3}} (3\sqrt{3} + 3\sqrt{3}) = 1.$$

$$\sigma^2 = \text{var}[X] = E[X^2] - \{E[X]\}^2 = 1 \text{ and } \mu = 0.$$

$\therefore$

By Chebyshev's inequality, we have

$$P\left[|X - \mu| \geq \frac{3}{2}\sigma\right] \leq \frac{\sigma^2}{c^2} = \frac{4}{9}.$$

$$\left(\because \sigma^2 = 1, c = \frac{3\sigma}{2}\right)$$

The actual probability is given by

$$P\left[|X - \mu| \geq \frac{3}{2}\sigma\right] = P\left[|X| \geq \frac{3}{2}\right]$$

$$(\because \mu = 0, \sigma = 1)$$

$$= 1 - P\left[|X| < \frac{3}{2}\right]$$

$$= 1 - P\left[-\frac{3}{2} < X < \frac{3}{2}\right]$$

$$= 1 - \int_{-3/2}^{3/2} \frac{dx}{2\sqrt{3}}$$

$$= 1 - \frac{1}{2\sqrt{3}} \left(\frac{3}{2} + \frac{3}{2}\right) = 1 - \frac{3}{2\sqrt{3}} = \frac{2 - \sqrt{3}}{2}$$

Hence, the actual probability is $(2 - \sqrt{3})/2$, whereas the upper bound obtained by Chebyshev's inequality is $4/9$.

Example 1.97. The rainfall X in a certain locality is a normally distributed random variable with mean 40 cm and variance 4 cm^2 . Find a simple upper bound on the probability that rainfall in a particular year will exceed the mean by 5 cm.

Solution. We have $\mu = 40$ and $\sigma^2 = 4$.

By Chebyshev's inequality, we have

$$P[|X - \mu| \geq c] \leq \sigma^2/c^2 \text{ or } P[|X - 40| \geq 5] \leq 4/25.$$

$$\therefore P[|X - 40| \geq 5] \leq 0.16.$$

The exact value of this probability is

$$P[|X - 40| \geq 5] = 2 P[X < 35]$$

$$= 2 P\left[\frac{X - 40}{2} < \frac{35 - 40}{2}\right] = \left[Z < -\frac{5}{2}\right],$$

$$\text{where } Z = \frac{1}{2} (X - 40) \sim N(0, 1)$$

$$= 2 \psi(-2.5) = 2 [1 - \psi(2.5)]$$

$$= 2[1 - .9938] = 2 \times .0062 = .0124.$$

Thus an upper bound to the probability is 0.16, which is totally different from the exact probability .0124.